

BDCBP401- CEMENT BASED BLOCKS AND PAVERS WORKS



Eng. MUGISHA Etienne

This module describes the knowledge, skills, and attitudes required to make cementbased blocks pavers. It is intended to be pursued by learner at TVET level IV in Building Construction. At the end of this module, the learner will be able to: select tools, equipment and materials, prepare mortar, produce cement blocks and cement pavers, prepare ground base, and lay pavers.

ASPEJ RWAMAGANA

RWAMAGANA - RWANDA

+250787160232

For calls & whatsapp

9/18/2023

LEARNING OUTCOME 1: SELECT TOOLS, EQUIPMENT AND MATERIALS

IC.1.1: IDENTIFICATION OF TOOLS, MATERIALS AND EQUIPMENT'S

Definition: tools and/or equipment are devices or instruments designed to perform specific tasks, make work more efficient, or achieve particular objectives.

CLASSIFICATION OF TOOLS AND EQUIPMENT'S WITH THEIR USES

- Personal protective equipment
- Setting out tools and equipment
- Cutting tools and equipment
- Mixing tools and equipment
- Placing tools and equipments
- Batching tools and equipment
- Cleaning tools and equipment
- Compaction tools and equipment
- Transporting tools and equipment
- Levelling tools and equipment

1. PPE (Personal Protective Equipment)

Refers to any equipment used to protect workers from exposure to variety hazards. Types of personal protective equipment include:

Helmet: is used in workplace environments such as industrial or construction sites to protect the head from injury due to falling objects, impact with other objects, debris, rain, and electric shock.

Safety boot: used to protect the foot from injury due sharp objects, Falling/crushing objects, hot/corrosive substances and wet /slippery surfaces.

Overall: is a safety clothing made of tough cotton, denim or linen and usually used as protective clothing while working, It is a loose fitting pair of pants with supporting cross-straps, or a full or half sleeve shirt that is worn over regular shirts, vests and trousers to protect them from heat, cold, splashes, sparks,

flames and flying debris etc. in the workplace.

Gloves: is used to protect hand from abrasion, sharp objects, hot object corrosive substances, toxic chemicals and high voltage equipment.

Dust Mask: is respiratory protection equipment used in environments with dusts encountered during construction or cleaning activities, such as dusts from drywall, brick, wood, fiberglass and silica.

Hearing protection: To provide protection during exposure to high and loud noise levels, you must wear safety earmuffs, and earplugs

Goggles: To provide protection during exposure to hazards like flying particles, metal, liquid chemicals, acidic liquids, light radiation, i.e., welding or cutting you must wear Safety goggles.



Dust mask



Gloves



Overall



Safety boot



Helmet

1. SETTING OUT, MEASURING AND LEVELING TOOLS AND EQUIPMENT

Settings out tools are essential for ensuring accurate placement and alignment of pavers and blocks during a construction or landscaping project. These tools help establish the layout, angles, and measurements for a precise installation. Here are some common setting out tools used for pavers and blocks:

- **String Lines:** String lines are used to create straight reference lines for alignment. They are stretched tightly between pegs or stakes to define the edges and layout of the installation area.
- **Spirit Level:** A spirit level, or bubble level, helps ensure that surfaces are level and even. It's crucial for maintaining a flat and uniform base for the pavers or blocks.

- **Tape Measure:** A tape measure is used to take accurate measurements for spacing, dimensions, and layout. It's essential for ensuring uniformity and precision in the installation.
- **Set Square:** A set square or carpenter's square helps establish right angles and 90-degree corners, ensuring that the pavers or blocks are laid out in a square or rectangular pattern.
- **Chalk Line:** Chalk lines are used to mark straight lines on the ground or other surfaces. They work by snapping a chalk-coated string onto the surface, creating visible guidelines.
- **Pegs or Stakes:** Wooden or metal pegs or stakes are driven into the ground to mark reference points, corners, and the edges of the installation area. String lines are often attached to these points.
- **Wooden Battens:** Wooden battens are straight boards used to create level lines or spacing guides. They are placed on the ground and used as a reference for laying pavers or blocks.
- **Marking Paint or Spray Paint:** These are used to mark the ground temporarily with lines or patterns for guidance. They can be easily removed after the installation is complete.
- **Corner Blocks or Templates:** These are used to ensure that corners and angles are consistent and well-defined during the installation

2. Batching tools and equipment

Batching tools are instruments and equipment used in various industries, particularly in manufacturing, construction, to measure, weigh, and combine specific quantities of raw materials or components to create batches of products with precise formulations. Here are some common types of batching tools:



Fig: gauge box

3. Mixing tools and equipment

Mixing tools are instruments and equipment designed to blend or combine

different substances or materials thoroughly.

The choice of mixing tool depends on the specific materials being mixed and the desired outcome. Here are some common types of mixing tools:

- **Concrete Mixers:** Concrete mixers are used to prepare the concrete mix by blending cement, aggregates, water, and additives. Different types of mixers are available, including drum mixers and pan mixers.
- **Spade**
- **Truck mixer**

4. Molding tools and equipment's

Molding tools and equipment are essential for creating and shaping pavers and blocks from raw materials like concrete or clay. Here are some common Molding tools and equipment used in pavers and blocks production:

- **Moulds:** These are the primary tools for shaping pavers and blocks. Moulds come in various sizes and shapes, allowing you to produce different designs and dimensions of pavers and blocks. They can be made from materials like steel, plastic, or rubber.
- **Paver and Block Presses:** These machines are used for automated or semi-automated production. They compress the concrete or clay mix into the moulds with high pressure, resulting in strong and durable pavers and blocks.
- **Paver and Block Machines:** These are larger and more complex machines used in industrial-scale production. They can handle multiple moulds simultaneously, increasing production efficiency.

5. Transportation and Handling Equipment

Transportation tools are those tools that are used for moving materials from one place to another. Examples of those tools and equipments include:

- **Wheelbarrows and carts:** Transport pavers or blocks to the installation area. And also used for transporting material to be used during manufacturing of pavers and blocks
- **Forklifts or skid-steer loaders:** Used for heavy lifting and transportation.

6. Compaction tools and equipment

- **Plate compactors:** Used to compact the base and bedding materials beneath pavers or blocks.
- **Vibratory rollers:** Larger machines for compacting larger areas.

- **Vibrating Table:** A vibrating table is used to settle and compact the concrete or clay mix inside the mould, ensuring that it fills all the crevices and corners evenly. This helps in achieving a dense and uniform product.

7. Laying and Placement Tools

- **Paver and block setters:** Assist in accurately placing pavers or blocks in the desired pattern.
- **Rubber mallets:** Used for tapping pavers or blocks into place.
- **Trowel:** is used to placing mortar into the mould
- **Steel float**

8. Cutting tools and equipment

Are tools or equipment that are used for cutting pavers and blocks

- **Paver saws:** Used to cut pavers and blocks to size.
- **Block splitters:** Designed to split blocks cleanly along the desired lines.

9. Cleaning and Maintenance Tools

Cleaning tools are instruments, equipment, and supplies designed to facilitate the cleaning and maintenance of various environments, surfaces, and objects to maintain cleanliness and hygiene.

- **Pressure washers:** Clean pavers and blocks after installation.
- **Sealant applicators:** Apply sealant to protect the surface.
- **sponge** is used to cleaning pavers and blocks
- **roller brushes:** used for cleaning on the working area
- **hand brush**

10. MAINTAINANCE OF TOOLS

Maintaining tools is important to ensure their longevity and optimal performance. Here are some general steps for tool maintenance:

- **Cleanliness:** Regularly clean your tools after use to remove dirt, debris, and any residue. Use a brush, rag, or compressed air to clean hard-to-reach areas.
- **Lubrication:** Apply lubricants or oils to moving parts to prevent rust and reduce friction. Be sure to use the appropriate lubricant for the type of tool.
- **Inspection:** Inspect your tools for signs of wear, damage, or loose parts. Replace or repair any damaged components promptly.
- **Sharpening:** Keep cutting tools sharp by regularly sharpening blades, bits, and edges. Dull tools can be dangerous and less efficient.

- **Storage:** Store your tools in a dry, clean, and organized space. Use toolboxes, racks, or pegboards to keep them accessible and protected from the elements.
- **Rust Prevention:** To prevent rust, store metal tools in a dry environment, and use rust inhibitors or coatings if necessary.
- **Calibration tools and equipments:** For precision tools, regularly calibrate them to maintain accuracy.
- **Handle Maintenance:** Check handles for cracks or splinters, and replace them if needed to ensure a secure grip and safety.
- **Safety:** Always follow safety guidelines when using tools, which include wearing appropriate personal protective equipment.
- **Manufacturer's Instructions:** Follow any specific maintenance recommendations provided by the tool's manufacturer.

NOTE: Remember that different tools may require unique maintenance procedures, so it's important to consult the user manuals or guidelines specific to each tool. Proper maintenance not only extends the life of your tools but also ensures your safety and the quality of your work.

1.1.2 IDENTIFICATION OF MATERIALS

The concrete commonly used to make concrete blocks and pavers is a mixture of powdered Portland cement, water, sand, and gravel.

These raw materials together make a light gray block having a fine surface texture and have high compressive strength.

The cement mortar also is commonly used to make different size and shape of pavers with a mixture of **cement, sand** and **water**

- The weight of a typical concrete block is **38-43 lb (17.2 to 19.5 kg)**. There is a higher percentage of sand and a lower percentage of gravel and water for the blocks. This made very dry, stiff mixture and hold shape when removing from the block mold.
- Concrete blocks are often made of **1:3:6 concrete** with a **maximum size aggregate of 10mm** or a cement-sand mixture with a ratio of **1:7, 1:8** or **1:9**. These mixtures, if properly cured, give concrete blocks compression strength well above what is required in a one-storey building.
- **There are two main types of concrete blocks** used in the construction industry: **hollow and solid**. Fully solid blocks are often used for projects like paving, where stability and durability are important

The difference between cement blocks and concrete blocks lies in some of the materials used to fabricate them. Cement blocks use fly ash aggregates, whereas concrete blocks use stone aggregates or fine crushed sand. Whilst cement blocks are lighter, concrete blocks are stronger.

DESCRIPTIONS OF TYPES AND PROPERTIES OF MATERIALS FOR CEMENTBASED BLOCKS AND PAVERS

Material is substance or mixture of substance that constitutes an object. Materials are usually categorized into two sources: natural and manmade.

Types of materials with their properties

1. Cement

Cement is a commonly used for binding materials in the construction of making blocks and pavers. The cement is obtained by burning a mixture of calcareous (calcium) and argillaceous (clay) material a very high temperature and then grinding clinker so produced to a fine powder

Properties of cement

➤ Strength

The compressive strength of hardened cement is the most important of all properties. Therefore, it is not surprising that the cement is always tested for its strength at the laboratory before the cement is used in important works. Strength tests are not made on neat cement paste because of difficulties of excessive shrinkage and subsequent cracking.

➤ Setting time

Setting times of cement: It is a term used to describe the stiffening of the cement paste.

- It is A period of 30 minutes as minimum setting time for initial setting and a maximum period of 600 minutes as maximum setting time is specified by IS code, provided the tests are conducted as per the procedure prescribed by IS 269-1967
- Setting of cement: It refers to a change from a fluid state to a rigid state. Although during setting, the cement paste acquires some strength. For practical purpose, it is important to distinguish setting from hardening which refers to the gain of strength of a cement paste.
- Initial setting time of cement paste: Is regarded as the time elapsed between the moments that the water is added to the cement, to the time that the paste starts losing its plasticity.
- Final setting time of cement paste: It is defined as the time interval between

the moment the water is added to the cement, and to the time when the cement has completely lost its plasticity and has gained sufficient firmness to resist certain definite pressure.

- **False set of a cement paste:** It is the abnormal premature stiffening of the cement within a few minutes of mixing with water. There is no noticeable heat evolved in this process

➤ **Soundness**

The cement when tested for soundness shall not have an expansion of more than 10mm. It is important that cement after setting does not undergo any noticeable change in volume. An excessive change in volume expansion of a cement paste after setting shows that the cement paste is unsound. It means that it is not suitable for the production of concrete.

The soundness of cement refers to the stability of the volume change in the process of setting and hardening. If the volume change is unstable after setting and hardening, the concrete structures will crack, which can affect the quality of buildings or even cause serious accidents, known as poor dimensional stability.

➤ **Consistency of Cement:**

The ability of cement paste to flow is consistency. It is measured by [Vicat Test](#). In Vicat test Cement paste of normal consistency is taken in the Vicat Apparatus. The plunger of the apparatus is brought down to touch the top surface of the cement.

- **Bulk density:** This is the mass of a unit volume of material in its natural state. Bulk density considers both solids and voids space, whereas relative density considers only mineral solids, the voids are eliminated.

TYPES OF CEMENT

In addition to ordinary Portland cement there are many varieties of cement that can be used for making blocks and pavers. Important varieties are briefly explained below:

- **White Cement:** The cement when made free from colouring oxides of iron, manganese and Chlorium results into white cement. In the manufacture of this cement, the oil fuel is used instead of coal for burning. White cement is used for the floor finishes, plastering, ornamental (decorative) works etc. In swimming pools white cement is used to replace glazed tiles. It is used for fixing marbles and glazed tiles.

- **Coloured Cement:** The cements of desired colours are produced by thoroughly mixing pigments with ordinary cement. The Chlorium oxide gives green colour. Cobalt produce blue colour.

Iron oxide with different proportion produce brown, red or yellow colour. Addition of manganese dioxide gives black or brown coloured cement. These cements are used for giving finishing touches to floors, walls, window sills, roofs etc.

- **Quick Setting Cement:** Quick setting cement is produced by reducing the percentage of gypsum and adding a small amount of aluminum sulphate during the manufacture of cement. Finer grinding also adds to quick setting property.

This cement starts setting within 5 minutes after adding water and becomes hard mass within 30 minutes. This cement is used to lay concrete under static or slowly running water.

- **Rapid Hardening Cement:** This cement can be produced by increasing lime content and burning at high temperature while manufacturing cement. Grinding to very fine is also necessary. Though the initial and final setting time of this cement is the same as that of Portland cement, it gains strength in early days. This property helps in earlier removal of form works and speed in construction activity.
- **Low Heat Cement:** In mass concrete works like construction of dams, heat produced due to hydration of cement will not get dispersed easily. This may give rise to cracks.

Hence in such constructions it is preferable to use low heat cement. This cement contains low percentage (5%) of tricalcium aluminates (C3A) and higher percentage (46%) of dicalcium silicate(C2S).

- **Pozzulana Cement:** Pozzulana is a volcanic power found in Italy. It can be processed from shales and certain types of clay also. In this cement pozzulana material is 10 to 30 per cent. It can resist action of sulphate. It releases less heat during setting.

It imparts higher degree of water tightness.

Its tensile strength is high but compressive strength is low. It is used for mass concrete works. It is also used in sewage line works.

TEST OF CEMENT FIELD METHOD

It is necessary to carry out field test of the cement sample before usage for making pavers and blocks on the sites to know if the cement is good or bad one.

You can carry out these field tests as follows:

- When you put your hand into a bag of cement you will receive a cool feeling.
- The color of ordinary Portland cement is powder greenish-grey.
- When you open a bag of cement, take careful look at the cement powder should not have any visible lumps.
- When you take a pinch of cement and feel it between your fingers, the cement should give a smooth feeling.
- When you take a sample of cement powder and throw it inside a bucket full of water, the cement particles float for sometimes before they sink into the bucket.

2. AGGREGATES

Aggregate is a granular or particle materials derived from the rock, such as sand, gravel, crushed stone concrete used with a hydraulic cementing medium to produce either concrete or mortar.

Classification of aggregate

They classified as **coarse** and **fine (sand)** aggregates based **on its size**.

Aggregate is used in two size groups:

Fine aggregate (sand): Particle size less than 4.75 mm but **Coarse aggregate:** Particle size more than 4.75 mm

- **Fine aggregate (Sand):** Sand consists of small grains of silica and is formed by the disintegration of rocks caused by weather action.
- **Coarse aggregate (gravels):** Coarse aggregate is a portion of aggregate retained on the 2.00 mm (No 10) sieve for bituminous concrete or retained on the 4.75 mm (No 4) sieve for Portland cement concrete

TYPES OF AGGREGATES:

■ **Natural aggregate:** they are extracted from the soil but without undergoing any modification in their internal structures.

■ **Artificial aggregate:** they also come from the soil but result is a mix of physical or chemical modification operated on them. This process of obtaining artificial aggregate can be done by crushing them into the small sizes needed on the site.

CLASSIFICATION OF AGGREGATE ACCORDING TO:

SIZE	SHAPE	TEXTURE	UNIT WEIGHT
Coarse	Rounded	Smooth	Normal
Medium	Irregular	Rough	Heavy
Fine	Angular	Granular	Light
Elongated		Crystalline	

Shape	
a.	 Rounded Water worn or shaped by attrition
b.	 Irregular Naturally irregular or partly shaped
c.	 Angular Well-defined
d.	 Flaky Thickness is small relative to other two dimensions
e.	 Elongated Length is larger relative to other two dimensions

Remark: Rounded aggregates give a more workable mix. Angular aggregates make concrete harder to place, work and compact but can make a strong concrete

PROPERTIES AND CHARACTERISTICS OF AGGREGATES

Aggregate should be:

- **Strong and hard:** a strong, harder aggregate will give a strong final concrete. Never use a crumble or flakey rock like sandstone.
- **Durable:** to resist wear, tear and weathering
- **Well graded:** aggregates should range in size so that they fit together well. This gives a strong and denser concrete.
- **Chemically inactive:** the aggregate don't react with the cement
- **Tough:** not easily broken or crushed
- **Clean:** dirt or clay sticking to the aggregates will weaken the bond between paste and aggregates.
 - It should be free from organic or vegetable matter.
 - It should be free from salt.
 - It should contain sharp, angular and coarse grains.
 - It should be free from coatings of clay and silt

SOURCES OF SAND

- **River sand**

It is obtained from the banks and beds of rivers. It may be fine or coarse. There are chances of the sand having silt and as such should be washed before use. Coarse sand is generally clean and is usually good for all purposes.

- **Sea sand**

It is collected from sea beach. It usually consists of fine rounded grains of brown colour. It usually contains salt which attracts moisture and causes disintegration of the work in which it is used. It can be used locally after it has been thoroughly washed to remove the salts.

- **Crushed stone.**

It is obtained by crushing stone to the particle size of sand. Sand found by crushing quality stone is usually an excellent fine aggregate.

3. Water

Water is an important ingredient of concrete as it actively participates in the chemical reaction with cement. Since it helps to form the strength giving cement gel, the quantity and the quality of water is required to be looked into very carefully.

The strength of concrete directly depends on the quantity and quality of water used in the mix

Qualities of good water for cement pavers and blocks

- It should be free from oils
- It should be free from acids or alkalis
- It should be free from Iron, Vegetables matter or other substance, which is likely to have adverse effect on concrete.
- It should be fit for drinking purpose

4. Admixture:

is defined as a material other than cement, aggregate and water used as an ingredient of concrete, added to the batch immediately before or during mixing, to modify the properties of ordinary concrete to make it more suitable for required situation.

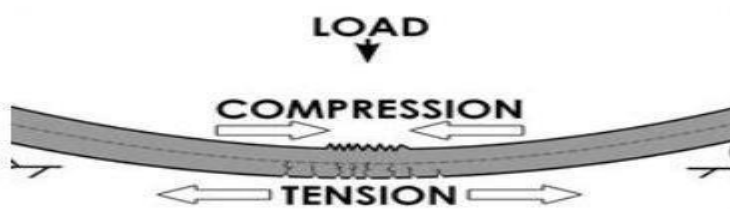
Instead of using a special cement, properties of cement can be changed by additives to cement.

Identification of factors for selecting cement blocks and pavers materials

➤ Strength

Strength is the most important parameter in selecting construction materials. It demonstrates the ability of the materials to withstand stresses such as compression, tension, and shear.

For instance, there are several steel grades with various yield strengths from which contractors can choose. If the building does not have to bear large loads, low steel grade can be used, which in this case, can reduce the cost. Nonetheless, the availability of steel in the market will play a significant role here.



➤ Durability

The durability of materials is important to construct a structure with a long life span without considerable maintenance cost.

➤ Materials Cost of pavers and blocks

The cost of construction materials will, in most cases, lead to the final decision regarding material selection.

Construction material costs comprise a large portion of the total cost, which is why it plays a critical role. Ultimately, it controls the entire cost of the project.

➤ Handling and Storage

Handling and storage of materials is another factor that must be considered because it affects construction time, requirements for labor and equipment for handling, and cost.

➤ Climate condition

Climate is another important factor that must be considered while choosing materials. Parameters such as average temperature range throughout the year, rain, snow, and weather, amount of sunlight, required ventilation, and wind are the areas of concern.

➤ **Availabilities of skilled labors**

When construction materials are selected, the decision-maker should know the extent of skills labor required to use that material. If skilled labor is needed, then it increases the cost of construction.

➤ **Sustainability**

The need for sustainable development is continuously increasing due to the demand for reducing carbon emissions.

Applicable codes and standards are continuously setting stricter specifications regarding carbon release. This leads decision-makers to lean toward sustainable materials like fly ash, ground granulated blast furnace slag, and silica fume, which are used to reduce cement utilization and reduce carbon dioxide emission. 9. Recycling

➤ **The Function of the Building**

The function of the building can govern the choice of construction materials because it directly affects the occupants. The selected material defines the space which is being constructed.

➤ **Aesthetic Appearance**

The type of material controls the appearance of the place. A specific appearance may be appealing to someone but not attractive to others.

➤ **Maintenance of that pavers and blocks materials**

Maintenance sometimes governs material selection because not only will good quality material need low cost to maintain, but it keeps the aesthetic appearance of the building.

➤ **Specifications**

Specifications are also an important material selection criterion. Material specifications and quality are established for a given project based on required strength, aesthetic appearance, durability, etc. .

LEARNING OUTCOME 2: FABRICATE CEMENT BLOCKS

2.1: QUALIFICATION OF CEMENT BLOCK MATERIALS

Introduction

Cement blocks, also known as concrete blocks or concrete masonry units (CMUs), are typically fabricated from a mixture of cement, aggregates (such as sand or gravel), and water. The process of making cement blocks involves several steps, including:

1. **Mixing:** Cement, sand, and aggregates are mixed in specific proportions to

create a homogenous mixture. The exact ratios can vary depending on the desired characteristics of the blocks.

2. **Molding:** The mixed concrete is placed into molds that have the shape and size of the desired blocks. These molds can be made of metal or wood and come in various configurations to produce different types of blocks, including standard concrete blocks, hollow blocks, or decorative blocks.
3. **Compacting:** To ensure the concrete is densely packed and free from voids, it is compacted within the molds. This can be done using a hydraulic press or a vibration process.
4. **Curing:** Once the blocks are molded and compacted, they need time to cure. Curing is the process of keeping the blocks in a controlled environment with proper temperature and humidity for a specific duration. This allows the concrete to harden and gain strength.
5. **Demolding:** After the curing period, the blocks are removed from the molds. At this point, they are still relatively soft, but they will continue to harden over time.
6. **Quality Control:** The blocks are inspected for quality to ensure they meet the required standards for strength, size, and appearance.
7. **Packing and Storage:** Once they pass quality control, the blocks are packed and stored until they are ready for distribution and use in construction projects.



Concrete block construction has gained importance and has become a valid alternative to fired clay bricks. The essential ingredients of concrete are cement, aggregate(sand, gravel) and water. Concrete blocks are produced in a large variety of shapes and sizes. They can be produced manually or with the help of machines.

Most commonly used concrete blocks are sized:

Length: 40cm (half blocks: 20cm) **Height:** 20cm

Width: 8/10/15/20cm

DIFFERENTS TYPES OF CEMENT BLOCK MATERIALS

- **Concrete Blocks:** Traditional concrete blocks are made from a mixture of Portland cement, aggregates (such as sand or crushed stone), and water. They are widely used in various construction applications due to their strength and durability. Concrete blocks can be solid or hollow, and their composition can be adjusted to achieve different levels of strength and insulation.
- **Hollow Blocks:** Hollow concrete blocks have voids or cavities inside them. These voids can be used to reduce the overall weight of the block and may also allow for the installation of reinforcing materials, insulation, or plumbing and electrical conduits. Hollow blocks are commonly used in load-bearing walls and partitions.
- **Lightweight Blocks:** Lightweight concrete blocks are made by using lightweight aggregates or additives, such as expanded clay, shale, or perlite. These blocks are lighter than standard concrete blocks and are often used in construction where weight reduction is important, such as in the construction of high-rise buildings.
- **Aerated Concrete Blocks:** Aerated concrete blocks, sometimes known as AAC (Autoclaved Aerated Concrete) blocks, are lightweight, precast concrete blocks that contain air bubbles. They are manufactured by adding aluminum powder to the concrete mix, which generates gas and creates a cellular structure. AAC blocks are known for their excellent insulation properties and are used in residential and commercial construction.
- **Decorative Blocks:** Decorative concrete blocks are made with various textures and designs on the surface. These blocks are often used for architectural and decorative purposes, such as building fences, garden walls, or exterior cladding. They come in a wide range of patterns and can resemble natural stone or other materials.
- **Split-Face Blocks:** Split-face concrete blocks have a rough, textured surface that is created by splitting the block after it has been molded and cured. These blocks are commonly used in architectural and landscaping applications for their aesthetic appeal.
- **Colored Blocks:** Concrete blocks can be manufactured with added pigments or dyes to achieve specific colors. Colored blocks are used to enhance the visual appeal of buildings and can be found in various shades to match different architectural styles.

- **Fire-rated Blocks:** These blocks are formulated to withstand high temperatures and are often used in applications where fire resistance is essential, such as in the construction of firewalls or fire-rated partitions.
- **Solid Blocks:** Solid concrete blocks are dense, heavy blocks with no internal voids. They are used in load-bearing applications, such as building foundations and structural walls, where strength and durability are paramount.

CONCRETE BLOCKS ARE CLASSIFIED INTO TWO MAIN TYPES:

- **Solid blocks:** have no cavities, or- according to US standards- have no voids amounting to not more than 25% of the gross cross-sectional area.
- **Hollow blocks:** are the most common type of concrete blocks, having one or more holes that are open at both sides. The total void area can amount to 50% of the gross cross-sectional area.

SPECIFICATION OF CEMENT BLOCKS MATERIALS

1. **Block Type:** Specify the type of block you need, such as standard concrete blocks, hollow blocks, lightweight blocks, aerated concrete blocks, decorative blocks, or any other specific type that suits your project requirements.
2. **Size and Dimensions:** Specify the size of the blocks in terms of length, width, and height. Common block sizes are typically 8 inches (200 mm) in height, 8 inches (200 mm) in depth, and either 16 inches (400 mm) or 8 inches (200 mm) in length. However, sizes can vary, and custom sizes are also available.
3. **Compressive Strength:** Compressive strength is a crucial specification. It indicates the maximum load in pounds per square inch (psi) that a block can withstand without breaking. Common compressive strength options range from 1,500 psi to 5,000 psi or more, depending on the application.
4. **Color:** If you require colored blocks, specify the desired color and whether it should be integral (mixed into the block) or surface-treated (applied after production).
5. **Texture and Finish:** Specify the desired texture or finish for the blocks, especially for decorative or architectural applications. Options may include smooth-faced, split-face, or other custom textures.
6. **Weight:** Consider the weight of the blocks, especially for applications where weight is a concern, such as in high-rise buildings. Lightweight blocks are available to reduce overall structural load.
7. **Thermal Insulation:** If you need blocks with enhanced thermal insulation properties, consider specifying the desired R-value or U-value, depending on your project's insulation requirements.

8. **Fire Rating:** If fire resistance is important for your project, specify the required fire rating, which may involve using specialized fire-rated blocks.
9. **Sound Insulation:** For applications where sound insulation is crucial, consider specifying the sound transmission class (STC) or noise reduction coefficient (NRC) of the blocks.
10. **Testing and Certification:** Ensure that the blocks meet local building code requirements and have the necessary certifications for quality and performance, such as ASTM (American Society for Testing and Materials) standards or equivalent standards in your region.
11. **Reinforcement Requirements:** Specify if the blocks need to be designed for the incorporation of reinforcing materials or other structural elements.
12. **Delivery and Handling:** Consider the logistics of transporting and handling the blocks on your construction site, and ensure that the blocks are manufactured and delivered in a manner that suits your project's requirements.
13. **Quantity:** Clearly state the required quantity of blocks for your project.
14. **Manufacturer or Supplier:** Specify the preferred manufacturer or supplier, or provide guidelines for selecting a qualified supplier if you have specific quality standards.

METHODS OF QUANTIFYING CEMENT BLOCK MATERIALS

Quantifying block materials is the process of calculating the quantity of materials requirement for making or construction of cement blocks

❖ Manual method

Quantifying or estimating the materials needed for a project involving cement blocks can be done manually by calculating the number of blocks, mortar, and other materials required. Here's a general method for manually quantifying cement blocks for a simple wall construction project:

1. Determine the Wall Dimensions: Measure the length, height, and thickness of the wall where the cement blocks will be used. Calculate the area of the wall in square feet or square meters.

2. Calculate the Number of Blocks:

- Determine the size of the blocks (length, height, and thickness).
- Calculate the area that one block will cover (length × height).
- Divide the total wall area by the area of one block to find the number of blocks required.

For example, if the wall area is 100 square feet, and each block covers 0.89

square feet, you would need approximately 112 blocks (100 sq ft / 0.89 sq ft per block).

3. Account for Mortar:

- Calculate the mortar thickness (usually around 3/8 to 1/2 inch or 10 to 12 mm) and adjust the block area calculation to include the mortar.
- For instance, if the mortar thickness is 0.5 inches and each block covers 0.89 square feet, you would need to add the mortar area for every block. Adjust the block size calculation accordingly.

4. Include Extra Blocks: It's advisable to purchase extra blocks to account for breakage, cutting, or any potential future repairs. A common rule of thumb is to add 5-10% extra blocks to the calculated quantity.

5. Calculate Mortar Quantity:

- Determine the volume of mortar required by multiplying the length, height, and thickness of the wall.
- Subtract the volume of the blocks from the total volume to find the mortar volume.
- Calculate the amount of dry mortar mix required and consider adding a small surplus to account for waste.

6. Estimate Other Materials: Depending on your project, you may need to estimate the quantity of other materials such as reinforcement, rebar, grout, and lintels.

❖ AUTOMATIC/MECHANICAL METHOD

An automatic method for quantifying cement block materials typically involves using construction estimating software or tools designed for this purpose. These software solutions can significantly streamline the process, providing accurate material quantities and cost estimates for your construction project. Here are the general steps for using automatic methods to quantify cement block materials:

1. Select Estimating Software: Choose a construction estimating software or tool that suits your needs. Many construction management software packages include estimating modules, and there are specialized estimating software options available as well.

2. Input Project Details:

Enter project specifications, such as wall dimensions, block sizes, and any special requirements.

Input the type and quantity of blocks you plan to use (e.g., standard blocks, lightweight blocks, or decorative blocks).

3. Calculate Wall Area: The software will automatically calculate the wall area based on your input dimensions.

4. Select Mortar Type: Specify the type of mortar you intend to use and its thickness.

5. Estimate Blocks: The software will automatically calculate the number of blocks required, taking into account the wall area, block size, and mortar thickness. It will also provide an option to add extra blocks for waste and adjustments.

6. Quantify Mortar: The software will determine the amount of mortar required, factoring in the specified thickness and wall area.

7. Include Additional Materials: If necessary, you can add other materials to the estimate, such as reinforcement, lintels, rebar, grout, and more.

8. Review and Refine: Review the software's estimate and make any necessary adjustments to the design or materials. You may need to refine your input to ensure the estimate aligns with the specifics of your project.

9. Generate Reports: Most estimating software allows you to generate detailed reports, including material quantities, costs, and other project data. These reports can be useful for project planning and procurement.

10. Consult Local Codes: Ensure that the materials and methods used in the estimate comply with local building codes and regulations

Technical Data	
Building element	Wall
Size	Various sizes
Mixing ratio	1 cement : 6-8 aggregate
Cement used per m ²	250 - 300kg
Resistance to earthquakes	Very good
Resistance to typhoons	Very good
Resistance to rain	Very good
Resistance to insects	Very good
Climatic suitability	All climates
Stage of experience	Widely used method
Production costs per m ²	Rp. 55,000
Durability	Approx. 30 years

❖ **Solid blocks are:**

- High compressive strength,
- resistance to weathering,
- impact and abrasion
- Capability of being moulded into components of any shape and size
- Good fire resistance up to about 400°C
- Rapid construction
- Very good stability

❖ **Hollow blocks are:**

- Can be made larger than solid blocks
- Are lighter in weight
- Construction of walls is easy and quick
- The voids can be filled with steel bars and concrete, achieving high earthquake resistance
- The air space provides good thermal insulation
- The cavities can be used for electrical installation and plumbing

➤ **Economic advantages of blocks as construction materials**

Production can be started with little capital.

Less working time required for brick laying work with concrete blocks. Less mortar consumption

Generally, production costs of concrete blocks are slightly lower than of fired bricks.

➤ **Raw Material**

- Cement:
- Ordinary Portland Cement
- Special cements Aggregate:
- Sand or gravel
- Max. particle size of coarser aggregates is 10mm.
- Suitable aggregates are usually obtained from natural sources (river beds, gravel pits, volcanic deposits) or from industrial by-processes (granulated blast furnace slags and red fly ash).

❖ **BATCHING OF CEMENT BLOCK MATERIAL**

Batching is the process of measuring and combining the ingredients required to make cement blocks in the correct proportions. The main ingredients for making cement blocks include cement, aggregates (such as sand or crushed

stone), water, and sometimes additives like plasticizers or colorants

➤ **Methods of batching**

- **Volume Batching:** In volume batching, the ingredients are measured by volume. This is commonly used in small-scale production where precision is not as critical. For example, you might use buckets or containers to measure the materials.

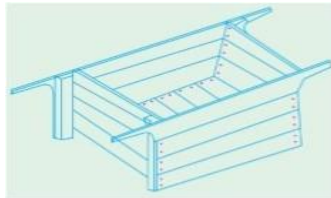


Figure: gouge box

- **Weight Batching:** Weight batching is a more precise method where the ingredients are measured by weight using scales or weigh batching equipment. This method is preferred in large-scale or commercial production to ensure consistency.



Figure:

❖ **Mixing of Cement Block Materials**

Once the ingredients are batched, they need to be thoroughly mixed to create a homogenous mixture that will form the cement blocks. This process involves determining the mix ratio, selecting a mixing method, and following specific procedures:

The mix ratio determines the proportions of each ingredient in the block mix. The mix ratio varies depending on the desired strength, density, and other properties of the blocks.

➤ **Mix ratio for blocks**

The mix ratio determines the proportions of each ingredient in the block mix. The mix ratio varies depending on the desired strength, density, and other properties of the blocks.

A common mix ratio for making standard concrete blocks is 1:6, which means one part cement to six parts aggregates (sand and/or crushed stone). The water content is adjusted to achieve the desired consistency.

- **Cement-Aggregate Ratio:**

Common ratios are 1:6 - 1:9 for cement _sand blocks and 1:3:6 for concrete blocks

- **Water-Cement Ratio:**

Only drinking quality water should be used to mix Concrete and recommended water-cement ratio is 0.5

➤ **METHOD OF MIXING**

There are two Methods of Mixing:

- **Hand Mixing:** This method involves manually combining the ingredients using shovels, hoes, or mixing troughs. It is suitable for small-scale or DIY projects but can be labor-intensive and less precise.

This method of mixing concrete is resorted to when the quantity of concrete to be used in a work is insufficient to warrant the necessity of machine.

This is used with advantage in places where machinery cannot be used on account of their non-availability or in works near a hospital where the noise of machine is not desirable. Hand mixing is done on a clean, hard and impermeable surface.

Cement and sand are first mixed dry with the help of shovels until the mixture attains a uniform colour. Aggregate are then added to this mixture and the whole mixture is then turned by shovels until the stone pieces uniformly spread throughout.

After this, desired quantity of water is poured into the heap from a can fitted with a rose. The mass is then turned until a workable mixture is obtained. It is advised to add 10% extra cement to guard against the possibility of inadequate mixing by this method.

- **Machine Mixing:** Machine mixing is the preferred method for large-scale production. It involves using a concrete mixer, which can be a drum mixer

or a pan mixer, to thoroughly blend the ingredients. Machine mixing ensures uniform distribution and consistent quality.

➤ **Procedures of Mixing:**

- **Dry Mixing:** In dry mixing, the cement and aggregates are first mixed together until they form a uniform dry mix. This step helps distribute the cement particles evenly through the aggregates.
- **Adding Water:** Water is added gradually to the dry mix while continuing to mix. The amount of water should be controlled to achieve the desired consistency, which is typically a workable but not overly wet mixture.
- **Mixing Time:** The mixture should be mixed for a sufficient duration to ensure complete homogenization. Mixing time can vary depending on the type of mixer and batch size but is typically several minutes.

❖ **POURING OF CEMENT MORTAR INTO CEMENT BLOCK MOULD**

Once the cement block mix is ready, it is poured into the block molds to shape and form the blocks. Here's a brief overview of the process:

- **Prepare the Molds:** Ensure the block molds are clean and properly lubricated or coated with a release agent to prevent sticking.
- **Fill in the Molds:** Pour the mixed cement mortar into the molds. Use a trowel or vibrating table to settle the mix and remove any air voids.

❖ **COMPACTION OF CEMENT BASED BLOCKS**

Compact the Mix: Depending on the type of mold and equipment used, you may need to apply pressure to compact the mix further. Vibrating tables or presses can be used for this purpose.

Level the Surface: Smooth and level the surface of the poured mix using a trowel or screed.

➤ **Advantages of Compaction:**

- **Improved Strength:** Compaction increases the density of the block, leading to improved compressive strength and overall structural integrity.
- **Durability:** Well-compacted blocks are less prone to cracking, chipping, and

damage over time, making them more durable.

- **Uniformity:** Compaction helps in achieving consistent block dimensions and reduces variations in size and shape.
- **Reduced Porosity:** It reduces the porosity of the blocks, making them less susceptible to moisture penetration, which can lead to degradation.
- **Better Bonding:** Proper compaction ensures good bonding between the cementitious materials, aggregates, and any reinforcing materials, resulting in a solid structure.
- **Energy Efficiency:** Compacted blocks often require less mortar during installation due to their uniformity, reducing material wastage and improving energy efficiency.

➤ **Compaction Methods for Cement-Based Blocks:**

There are two main methods of compaction for cement based blocks:

- Manual compaction
- Mechanical compaction.

1. Manual Method: the way of compacting by using hand tools

- **Hand Tamping:** In this method, laborers use hand tools like tamping bars or mallets to compact the block mixture within the mold. It requires physical effort and can be time-consuming, but it is suitable for small-scale production or where machinery is unavailable.
- **Hand Compressors:** Hand-operated block-making machines with a manual lever or handle can be used to compress the block mixture within the mold. This method increases productivity compared to hand tamping.

2. Mechanical Method: using mechanical energies

- **Vibrating Compactors:** Vibrating machines are commonly used for compacting cement-based blocks. They consist of a vibrating table or platform on which the block molds are placed.

The vibrations help settle the mixture and remove air voids, resulting in tightly compacted blocks. Vibrating compactors are highly efficient and suitable for large-scale production.

- **Hydraulic Press Machines:** Hydraulic presses exert immense pressure on the block mixture to compact it. These machines can produce high-density blocks with excellent strength and uniformity. They are often used for

producing high-quality, load-bearing blocks.

- **Pneumatic Compactors:** Pneumatic compactors use air pressure to compress the block mixture. They are suitable for various block sizes and shapes and are known for their precision and consistency.

The choice between **manual** and **mechanical compaction methods** depends on factors such as **production scale**, **budget**, and **desired block quality**. While manual methods are suitable for small-scale or DIY projects, mechanical methods are essential for large-scale production and achieving consistent, high-quality cement based blocks.

➤ **Process of compaction**

Put the mixture into wooden or steel mould boxes or Moulding machine. De-mould immediately after compaction

In details'

- Prepare first the head layer:
 - Fill the mix into prepared tile moulds up to fixed level to obtain uniform thickness of first layer
 - Secondly, prepare the body layer:
 - Fill the prepared mix in stages on top of the first layer until the mould is completely filled
- Compaction of cement-based blocks is an essential process in their production to achieve the desired density, strength, and durability. Compaction involves reducing the voids and air pockets in the mixture and ensuring that the cement-based material is tightly packed together.

➤ **DE-MOULDING OF CEMENT BASE BLOCKS**

De-Moulding: is the process of removing blocks into the formwork

➤ **CURING FABRICATED CEMENT BLOCKS**

Curing is a critical step in the production of fabricated cement blocks as it ensures the blocks gain strength and durability by allowing them to properly hydrate and set. The curing process involves maintaining adequate moisture and temperature conditions for a specified period. In Rwanda, as in many other places, curing standards are in place to ensure the quality of concrete blocks. Here's how to apply a cement block curing process complying with Rwanda standards:

➤ **Period of Curing:**

In Rwanda, the curing period for cement blocks is typically 7 days. However, this duration may vary based on factors like block composition, ambient temperature, and block size. It is important to consult the specific standards or guidelines provided by Rwandan authorities or relevant organizations for the most accurate information.

➤ **Identification of Drying Types**

There are two main types of drying during the curing process:

- Artificial Drying:
- Natural drying

Artificial drying involves the use of external methods to accelerate the curing process. It may include covering the blocks with wet burlap or plastic sheets, as well as the use of curing compounds or curing chambers. In some cases, artificial heating or steam curing can be used, but this should be done cautiously to avoid rapid temperature changes that can lead to cracking.

Natural drying relies on environmental conditions to cure the blocks. It typically involves keeping the blocks moist by spraying or sprinkling water on them regularly and covering them to prevent rapid moisture loss.

Natural drying is a slower process compared to artificial drying, but it is often more energy-efficient and suitable for small-scale block production.

➤ **Application of Cement Block Curing Process Complying with Rwanda Standards:**

To comply with Rwanda standards for cement block curing, follow these steps:

1. **Cover the Blocks:** After demoulding, cover the blocks with wet burlap, plastic sheets, or other suitable materials to retain moisture.
2. **Maintain Moisture:** Keep the blocks moist throughout the curing period. This can be achieved by regular wetting or sprinkling. The blocks should never be allowed to dry out during curing.
3. **Control Temperature:** Maintain the temperature within the recommended range. Extremes of temperature can affect the curing process. Adequate insulation can help moderate temperature fluctuations.
4. **Duration:** Follow the specified curing period, which is typically 7 days in Rwanda, but can vary based on factors like block composition and size.
5. **Record Keeping:** Maintain records of the curing process, including dates, methods used, and environmental conditions. This documentation can be

important for quality control and compliance with standards.

6. **Quality Control:** Regularly inspect the blocks during the curing process for any defects, and take corrective measures as needed.
7. **Compliance with Local Standards:** Always ensure that your curing process aligns with the specific standards and guidelines established by Rwandan authorities or relevant organizations. These standards may be subject to updates or revisions, so it's important to stay informed.

Adhering to proper curing practices in line with Rwanda's standards is crucial to produce high-quality, durable cement blocks that meet structural and safety requirements. It helps ensure that the blocks reach their intended strength and durability for construction applications.

Storing of block cement:

- Do not expose to direct sun light; keep the blocks in a dry and covered area.
- Store for 2 weeks before usage.
- Demould blocks immediately after compaction

Concrete Blocks Walls Estimation

There are various methods for calculating the number of blocks in the wall. In this article, we will see two simple methods for calculating the number of blocks used in the wall.

- Surface Area
- Method Volume Method

How to calculate the number of Blocks by Surface Area Method?

For calculating the number of blocks, you have to know:

- The surface area of the wall
- Surface area of the standard concrete block
- The surface area of openings in the wall

a) Surface Area Method

Step 1. The Length & Height of the Wall

Step 2. Now calculate the surface area of the wall
 $\text{Surface Area} = \text{Length} \times \text{Height}$

Step 3. Now calculate the Surface Area of Openings The openings like a door, windows, etc.

Step 4. Now subtract the openings from the surface area of the Wall

Step 5. Now calculate the surface area of concrete block with mortar Use The standard block dimensions and mortar allowance is “1”

Step 6. Now divide the total surface area of the wall by the surface area of one block
 $\text{Number of blocks} = \frac{\text{surface area of Wall}}{\text{Surface area of the block}}$ here consider 5% wastage of concrete blocks

Total number of blocks required = calculated blocks + wastages

b) Calculate by Volume Method

To calculate the number of pavers by volume method you need to have:

- The volume of the wall
- Volume of the standard concrete block
- The volume of the openings in the wall

Step 1. First determine the Width, Height, and Thickness of the Wall

Step 2. Now the calculate the volume of the wall
 $\text{Volume} = \text{Width} \times \text{Height} \times \text{Thickness}$

Step 3. The volume of the Openings, The openings like a door, windows, etc.,

Step 4. Subtract the volume of the Openings from Volume of Wall

Step 5. Now calculate the volume of the concrete block with the Mortar allowance

Step 6. Divide the total volume of a wall by volume of one block
 $\text{Number of blocks} = \frac{\text{Volume of Wall}}{\text{Volume of Block}}$ and consider 5% -10% wastage of concrete blocks

Note that if there are no openings in the wall, skip step 4.

Example:

Estimate the amount of Concrete blocks to build a wall of 28m and a height of 2.5 m, has a door with dimensions of 2.5 * 1.0 m, and window of dimensions 1.5 * 2.5 m.

By using area method

To estimate the number of concrete blocks needed to build a wall, you can use the area method. First, calculate the total area of the wall and then subtract the areas of the door and window to get the net wall area that needs to be covered with concrete blocks.

Calculate the total wall area:

Total Wall Area = Length × Height

Total Wall Area = 28m × 2.5m = 70 square meters

Calculate the area of the door:

Door Area = Width × Height

Door Area = 2.5m × 1.0m = 2.5 square meters

Calculate the area of the window:

Window Area = Width × Height

Window Area = 1.5m × 2.5m = **3.75 square meters**

Calculate the net wall area:

Net Wall Area = Total Wall Area - (Door Area + Window Area)

Net Wall Area = 70 square meters - (2.5 square meters + 3.75 square meters)

Net Wall Area = 70 square meters - 6.25 square meters

Net Wall Area = **63.75 square meters**

Now that you have the net wall area that needs to be covered with concrete blocks, you can calculate the number of blocks needed. The number of blocks required will depend on the size of the concrete blocks you plan to use. You'll need to know the dimensions of the concrete blocks in order to proceed with the calculation.

Let's assume that each concrete block has dimensions of 0.2m x 0.4m (which are common sizes). To find the number of blocks needed:

Calculate the area of one concrete block:

Block Area = Block Width × Block Height
Block Area = 0.2m × 0.4m = **0.08 square meters**

Calculate the number of blocks needed:

Number of Blocks = Net Wall Area / Block Area

Number of Blocks = 63.75 square meters / 0.08 square meters/block
Number of Blocks ≈ **796.88 blocks**

You would need approximately **797 concrete blocks** to build the wall of 28m in length and 2.5m in height, taking into account the door and window openings. Keep in mind that this is an estimate, and the actual number of blocks required may vary based on factors such as mortar thickness and block size. It's a good practice to purchase a little extra to account for any waste or unexpected needs.

LEARNING OUTCOME 3: FABRICATE CEMENT PAVERS

3.1: QUALIFICATION OF CEMENT PAVERS MATERIALS

31.1. Introduction

Cement pavers, also known as concrete pavers, are precast concrete blocks or tiles used in outdoor hardscaping projects for creating paths, driveways, patios, and other paved surfaces. They are a popular choice for their durability, versatility, and ease of installation.

These pavers are typically made by mixing cement, sand, gravel, and water, and then molding the mixture into specific shapes and sizes. The manufacturing process involves curing the concrete to ensure strength and durability. Cement pavers come in various colors, textures, and patterns, providing homeowners and landscapers with a wide range of design options to complement different architectural styles.



Materials Used in Cement Pavers:

- **Cement:** Acts as the primary binding agent that holds the paver together. Typically, Portland cement is used.
- **Sand:** Fine aggregate that provides bulk and strength. Helps in binding the other materials and gives the paver its texture.
- **Water:** Activates the cement, initiating the chemical process of hydration that hardens the mixture. The amount of water affects the paver's strength and durability.
- **Aggregates:** Typically crushed stone or gravel is used as coarse aggregates. These provide the necessary bulk and strength to the paver.
- **Admixtures (Optional):** Chemicals added to improve certain properties like workability, curing time, or durability. Can also be used to make the pavers more resistant to environmental factors.

Uses of Cement Pavers:

1. **Driveways:** Due to their durability and ability to withstand heavy loads, cement pavers are commonly used for driveways. They provide a stable and attractive surface for vehicles.
2. **Patios and Walkways:** Cement pavers are used to create patios, walkways, and garden paths. Their versatility in shapes, colors, and sizes makes them ideal for creating visually appealing outdoor spaces.
3. **Pool Decks:** The slip-resistant surface of pavers makes them a popular choice for pool decks. They are also easy to maintain and can handle water exposure well.
4. **Commercial Landscaping:** Used in parks, plazas, and other public spaces. The durability and customizable nature make them ideal for high-traffic areas.
5. **Roadways:** Interlocking cement pavers are used in the construction of roadways, particularly in areas that require permeable surfaces to manage storm water.
6. **Retaining Walls:** Though typically not the first choice, some cement pavers can be used in constructing low retaining walls.
7. **Erosion Control:** Cement pavers are used in areas prone to erosion, like slopes or embankments, to provide stability and prevent soil loss.

The advantages of cement pavers include

- **Durability:** Concrete is a durable material, and properly installed cement pavers can withstand heavy foottraffic and vehicular loads.
- **Versatility:** They are available in various shapes, sizes, and colors,

allowing for creative and customizable designs.

- **Ease of installation:** Cement pavers are relatively easy to install, especially in comparison to poured concrete or asphalt surfaces. They can be laid on a sand or gravel base without the need for mortar.
- **Repairability:** If one or more pavers become damaged, they can be easily replaced without disturbing the entire surface.
- **Cost effective:** cement pavers are often more cost-effective than other paving materials, making them a popular choice for budget-conscious projects.
- **Low maintenance:** Once installed, cement pavers require minimal maintenance. Regular cleaning and occasional sealing can help prolong their life and maintain their appearance.

Cement pavers are a versatile and practical choice for various outdoor applications, providing an aesthetically pleasing and long-lasting solution for landscaping and landscaping projects.

DIFFERENTS TYPES OF CEMENT BLOCK MATERIALS

There are several types of cement pavers, each with its own characteristics and intended use. The choice of a specific type of cement paver often depends on the application, design preferences, and the level of durability required. Here are some common types of cement pavers:

- **Interlocking concrete pavers:** These pavers have unique shapes and edges that allow them to fit together like pieces of a puzzle, creating a strong and cohesive surface. Interlocking pavers are commonly used for driveways, walkways, and patios.
- **Traditional concrete blocks:** Traditional or square concrete pavers are basic square or rectangular blocks used for various applications.
- **Tumbled concrete pavers:** Tumbled pavers have rounded edges and a weathered appearance, giving them a more natural and aged look.
- **Permeable concrete pavers:** are designed to allow water to pass through the surface, promoting drainage and reducing water runoff. These are environmentally friendly and are often used in areas where storm water management is a concern.
- **Slip resistant pavers:** These pavers have textured surfaces to provide better traction, making them suitable for areas prone to wet conditions, such as pool decks or walkways in rainy climates.

- **Stamped concrete pavers:** are designed to mimic the appearance of natural materials, such as stone or brick. They are created by pressing a pattern into the wet concrete before it sets, resulting in a textured and decorative surface.
- **Hallow concrete pavers:** These pavers have a hollow core, reducing their weight and making them easier to handle during installation. They are often used in areas where weight is a concern, such as rooftop gardens.
- **Segment retaining wall (SRW) pavers:** These pavers are designed for use in retaining walls, allowing the construction of walls with an attractive and durable facing.
- **Porcelain pavers:** While not made of traditional concrete, porcelain pavers are worth mentioning. These pavers are made from high-fired clay and can resemble natural stone or wood. They are known for their durability and resistance to staining.

Requirements to select cement pavers in short sentences

- **Load-Bearing Capacity:** Ensure pavers can withstand the expected weight and traffic.
- **Durability:** Choose pavers that resist wear, weathering, and environmental conditions.
- **Slip Resistance:** Select pavers with a textured surface for safety, especially in wet areas.
- **Aesthetic Appeal:** Consider color, shape, and size to match the design and style of the project.
- **Permeability:** Opt for permeable pavers if drainage and stormwater management are concerns.
- **Ease of Installation:** Choose pavers that fit well together and are easy to install and maintain.
- **Budget:** Select pavers that offer the best value within the project's budget constraints.

QUANTIFY CEMENT PAVERS MATERIALS

To **quantify cement paver materials** means to calculate or determine the exact amount or quantity of each material needed to produce or install cement pavers for a specific project.

Function of quantifying cement pavers materials are:

- Provides accurate budget forecasting.
- Ensures sufficient material supply.

- Minimizes excess material and waste.
- Helps avoid delays due to material shortages.
- Ensures consistent material usage for uniformity.

Methods of quantify cement pavers materials

❖ Manual method

Quantifying materials for a paver project manually involves a series of calculations based on the dimensions of the project area and the specifications for the pavers and base materials. Here's a step-by-step guide for the manual method of quantifying paver materials:

- 1. Measure the Area:** Measure the length and width of the area where the pavers will be installed. Multiply these dimensions to determine the square footage (length x width = area).
- 2. Determine Paver Thickness:** Decide on the thickness of the paver bed. Common thicknesses for paver projects range from 2 to 4 inches.
- 3. Calculate Base Material Volume:** Determine the type of base material to be used (e.g., crushed stone, gravel). Calculate the volume of base material needed using the formula: $\text{Volume} = \text{Area} \times \text{Thickness}$
- 4. Estimate Sand Bed Volume:** Determine the volume of sand needed for the bedding layer beneath the pavers. This is typically a thin layer, and the volume can be calculated using the same formula as for the base material.
- 5. Calculate Cement Quantity for Mortar:** Determine the ratio of cement to sand in the mortar mix. A common ratio is 1:3 (one part cement to three parts sand). Calculate the volume of cement needed using this ratio and the volume of sand:
- 6. Account for Waste:** Factor in a percentage for waste, spillage, and breakage. A common allowance is 5-10% of the calculated quantities.
- 7. Calculate the Number of Pavers:** Divide the total square footage of the area by the size of an individual paver to determine the number of pavers needed.
- 8. Consider Edge Restraints:** If using edge restraints, calculate the linear footage required around the perimeter of the paved area.
- 9. Account for Joint Sand:** If joint sand will be used between the pavers, estimate the quantity based on the size of the joints.
- 10. Summarize Quantities:** Summarize the quantities of base material, sand, cement, pavers, and any other materials needed for the project.

11. Verify with Local Codes: Check local building codes and regulations to ensure compliance with any specific requirements for base thickness, materials, and installation methods.

❖ **Automatic/mechanical method**

An automatic or digital method of quantifying cement paver materials involves using specialized **software or online calculators designed for construction and landscaping projects**. These tools can streamline the process and provide more accurate estimates. Here's a general guide on how to use such tools:

1. Online Calculators or Software: Many websites and software applications are available that provide automated calculations for paver projects. Look for tools specifically designed for estimating materials for paving projects.

2. Input Project Details: Enter the project details into the calculator or software. This typically includes the dimensions of the area where the pavers will be installed, the type and size of the pavers, and any additional parameters such as base thickness.

3. Specify Paver Type and Size: Select the type of pavers you plan to use (interlocking, traditional, etc.) and input the dimensions of the pavers. Some tools may have a database of common paver sizes and types.

4. Choose Base and Bedding Materials: Specify the type and thickness of the base and bedding materials. The tool may provide options for different materials commonly used in paver installations.

5. Customization Options: Some calculators allow for customization based on specific project needs, such as the inclusion of edge restraints, joint sand, or adjustments for waste.

6. Receive Quantities and Recommendations: After inputting all necessary details, the calculator or software will provide you with quantities of materials required, including the amount of pavers, base material, sand, and cement. It may also offer recommendations for installation.

7. Print or Save Results: Some tools allow you to print or save the results for future reference. This can be useful when purchasing materials or sharing the information with contractors.

❖ **Verify with Professional Advice:** While automated tools are helpful, it's advisable to verify the results with professional advice or consult with the paver manufacturer for any specific recommendations.

❖ **BATCHING OF CEMENT PAVERS MATERIALS**

Batching cement paver materials is the process of precisely measuring and

mixing the components needed to produce consistent and quality pavers.

METHOD OF BATCHING

- **Volume Batching:** In volume batching, the ingredients are measured by volume. This is commonly used in small-scale production where precision is not as critical. For example, you might use gauge box of known volume to measure the materials

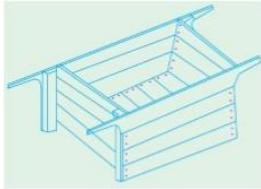


Fig: gauge box

- **Weight Batching:** Weight batching is a more precise method where the ingredients are measured by weight using scales or weigh batching equipment. This method is preferred in large-scale or commercial production to ensure consistency.



Fig: weight batcher machine

MIXING OF CEMENT PAVERS MATERIALS

Once the ingredients are batched, they need to be thoroughly mixed to create a homogenous mixture that will form the cement blocks. This process involves determining the mix ratio, selecting a mixing method, and following specific procedures:

Mix ratio for pavers

The mix ratio determines the proportions of each ingredient in the block mix. The mix ratio varies depending on the desired strength, density, and other

properties of the blocks.

Here are a few common mix ratios used for cement pavers:

- 1. 1:3:3 mix ratio:** This mix consists of one-part cement, three parts sand, and three parts coarse aggregate. This ratio is often used for making high-strength concrete pavers.
- 2. 1:2:4 mix ratio:** In this mix, there is one-part cement, two parts sand, and four parts coarse aggregate. This mix provides a good balance of strength and workability and is commonly used for various types of concrete applications, including pavers.
- 3. 1:1.5:3 mix ratio:** This mix includes one-part cement, 1.5 parts sand, and three parts coarse aggregate. It is often used for making concrete pavers where a smoother finish is desired.
- 4. 1:2.5:3.5 mix ratio:** With one-part cement, 2.5 parts sand, and 3.5 parts coarse aggregate, this mix is suitable for creating concrete pavers with a good balance of strength and workability.
- 5. 1:4:6 mix ratio:** In this mix, there is one-part cement, four parts sand, and six parts coarse aggregate. It is commonly used for larger concrete applications where a higher proportion of aggregate is acceptable.

The water-cement ratio is a crucial factor in concrete mix design, including for cement pavers. It refers to the proportion of water, by weight, to the weight of cement used in the concrete mix. **The water-cement ratio has** a significant impact on the **workability, strength, and durability of the concrete**. The goal is to find a balance: enough water for workability, but not so much that it compromises the strength and durability of the final product.

For cement pavers, a typical water-cement ratio ranges between **0.4 and 0.6**. However, the exact ratio can vary based on factors such as the type of paver, mix design, and environmental conditions. Here are some general guidelines:



METHOD OF MIXING

There are two Methods of Mixing:

- ❖ **Hand Mixing:** This method involves manually combining the ingredients using shovels, hoes, or mixing troughs. It is suitable for small-scale or DIY projects but can be labor-intensive and less precise.

- This method of mixing concrete is resorted to when the quantity of concrete to be used in a work is insufficient to warrant the necessity of machine.
 - This is used with advantage in places where machinery cannot be used on account of their non-availability or in works near a hospital where the noise of machine is not desirable. Hand mixing is done on a clean, hard and impermeable surface.
 - Cement and sand are first mixed dry with the help of shovels until the mixture attains a uniform colour. Aggregative are then added to this mixture and the whole mixture is then turned by shovels until the stone pieces uniformly spread throughout.
 - After this, desired quantity of water is poured into the heap from a can fitted with a rose. The mass is then turned until a workable mixture is obtained. It is advised to add 10% extra cement to guard against the possibility of inadequate mixing by this method.
- ❖ **Machine Mixing:** Machine mixing is the preferred method for large-scale production. It involves using a concrete mixer, which can be a drum mixer or a pan mixer, to thoroughly blend the ingredients. Machine mixing ensures uniform distribution and consistent quality.

Procedures of machine mixing:

1. Start or turn on the mixer
2. Put in about 2/3 of the water you need
3. Add the sand and then the cement
4. Mix together until it all looks the same color
5. Add the rest of the water a bit at a time until you get a mix that is workable but not too wet. Use the least water you can
6. Mix for another 2 minutes Pour out and make the next batch.
7. Don't forget to wash out the mixer as soon as you have finished.

POURING OF CEMENT MORTAR

Pouring cement mortar is a process commonly used in construction to create a solid, durable surface or structure.

Process of pouring cement mortar into cement pavers mould

Pouring cement mortar into cement paver molds is a process used to create custom concrete pavers for driveways, walkways, patios, and other outdoor spaces.

Steps to Pour Cement Mortar into Paver Molds:

1. Prepare the Work Area:

- **Clean the Area:** Ensure the workspace is clean and free of debris
- **Gather Materials and Tools:** You'll need paver molds, cement, sand, water, mixing tools (bucket, shovel, or mechanical mixer), a trowel, and a level.

2. Prepare the Cement Mortar:

- **Mix Ingredients:** Combine cement, sand, and water in the appropriate ratio. A typical mix might be **1 part cement to 2-3 parts sand**, depending on the desired strength and texture of the pavers.
- **Add Water Gradually:** Add water slowly while mixing to achieve a consistent, workable mortar. The mixture should be thick enough to hold its shape but not too dry.

3. Prepare the Paver Molds:

- **Clean the Molds:** Ensure that the molds are clean and free of any old mortar or debris. This will ensure a smooth finish for the pavers.
- **Apply Release Agent:** If desired, apply a thin layer of a release agent (such as oil or a specialized mold release spray) inside the molds. This will make it easier to remove the pavers once they are set.

4. Pour the Cement Mortar into the Molds:

- **Fill the Molds:** Pour the prepared cement mortar into the molds. Use a trowel or a similar tool to spread the mortar evenly and fill all corners and edges of the mold.
- **Tap the Molds:** Gently tap the sides of the molds to remove any air bubbles and ensure that the mortar settles evenly. This step is important for achieving a smooth, uniform surface.

5. Level and Smooth the Surface:

- **Leveling:** Use a straight edge or a trowel to level the top of the mortar, ensuring it is even with the edges of the mold.
- **Smoothing:** Smooth the surface with a trowel to create a flat, even finish. You can add texture or patterns at this stage if desired.

6. Curing the Pavers:

- **Initial Setting:** Allow the pavers to begin setting. This usually takes several hours, depending on the temperature and humidity.
- **Moisture Maintenance:** Keep the pavers moist during the curing process by covering them with a damp cloth or plastic sheet. This helps prevent the mortar from drying too quickly and cracking.
- **Final Curing:** The pavers should be allowed to cure for at least 24-48 hours before removing them from the molds. Full curing may take several days, depending on the weather conditions.

7. Demolding the Pavers:

- **Removing from Molds:** Carefully remove the pavers from the molds by gently tapping the mold and pulling the paver out. Be careful not to damage the edges or surface.
- **Final Curing:** Place the pavers in a shaded area and allow them to cure completely for an additional few days. Keep them moist during this time to ensure full strength development.

8. Finishing Touches:

- **Edge Smoothing:** If necessary, smooth the edges of the pavers with a file or sandpaper.
- **Cleaning:** Clean the molds and tools for future use.

9. Installation:

- Once fully cured, the pavers can be installed in the desired location.

COMPACTION OF CEMENT PAVERS

Compaction of cement pavers refers to the process of densifying and consolidating the cement mortar or concrete mixture used in making pavers to eliminate air voids and ensure a uniform, solid structure.

Advantages of compaction

- **Increases Strength:** Compaction enhances the compressive strength of cement pavers.
- **Reduces Voids:** It minimizes air pockets, resulting in a denser, more solid structure.
- **Improves Durability:** Compacted pavers resist wear and weathering more effectively.
- **Enhances Water Resistance:** Lower porosity reduces water absorption.

and freeze-thaw damage.

- **Ensures Uniformity:** Compaction provides a consistent surface finish and even density.

❖ **Compaction Methods for Cement-Based pavers**

There are two main methods of compaction for cement based pavers:

- Manual compaction
- Mechanical compaction.

A. Manual compaction

Manual compaction is a method used in construction where the process of densifying and consolidating a material, such as cement mortar, is done manually using hand tools.

Key Aspects of Manual Compaction:

- **Tools Used:** Common tools include trowels, tamping rods, mallets, or hand-held compactors or compressors.
- **Process:** The worker presses, taps, or tamps the material to compact it, ensuring that it fills all voids and achieves the desired density.

Equipements used:

- **Hand Tamping:** In this method, laborers use hand tools like tamping bars or mallets to compact the pavers mixture within the mold. It requires physical effort and can be time-consuming, but it is suitable for small-scale production or where machinery is unavailable.
- **Hand Compressors:** Hand-operated block-making machines with a manual lever or handle can be used to compress the paver mixture within the mold. This method increases productivity compared to hand tamping.

B. Mechanical compaction

Mechanical compaction is a method used in construction where machinery or mechanical devices are employed to densify and consolidate materials like cement mortar or concrete.

Key Aspects of Mechanical Compaction:

- **Tools Used:** Equipment like **vibrating tables**, **mechanical vibrators**, or **hydraulic presses** are commonly used.

- **Process:** The machinery applies consistent force, pressure, or vibration to the material, ensuring it fills the mold completely and achieves a uniform density.

Compaction of cement paver's process

The compaction process for cement pavers involves **filling molds with cement mortar**, then **using manual or mechanical methods to eliminate air pockets** and **ensure even density**. This is achieved by either hand tamping (manual compaction) or applying vibrations or pressure (mechanical compaction). Proper compaction enhances the strength, durability, and uniformity of the pavers.

By using steps:

1. **Pour Mortar:** Fill the paver mold with cement mortar.
2. **Level Surface:** Smooth and level the mortar within the mold.
3. **Manual or Mechanical Compaction:** Use hand tools or machinery to compact the mortar, eliminating air pockets.
4. **Vibration/Pressing:** Apply vibrations or pressure (mechanically) for uniform density.
5. **Final Smoothing:** Ensure the surface is even and smooth after compaction.
6. **Curing:** Allow the paver to set and cure properly before Demolding.

❖ DE-MOULDING OF CEMENT BASE BLOCKS

De-molding, also known as demolding or demoulding, is the process of removing cured or set concrete pavers from their molds once they have reached the appropriate level of strength and have undergone the curing period.

Demolding Process for Cement Pavers:

1. **Complete Curing:** Ensure pavers have finished the recommended curing period for optimal strength and durability.
2. **Prepare the Area:** Set up a clean, level workspace with enough room for handling pavers.
3. **Inspect Molds:** Check for damage or irregularities and confirm that pavers are fully cured.
4. **Check Release Agent:** If used, ensure the release agent is still effective for smooth removal.
5. **Loosen Edges:** Gently tap the mold sides with a rubber mallet to loosen the paver edges, starting around the perimeter.

6. **Lift Pavers:** Carefully remove the pavers, tilting the mold if necessary to help them slide out smoothly.
7. **Handle Sticking Pavers:** If a paver sticks, avoid force. Tap gently or use air pressure to aid in release.
8. **Inspect Pavers:** Check for imperfections or damage, ensuring they meet quality standards.
9. **Clean Molds:** Thoroughly clean molds before reuse, removing any concrete residue or release agent.
10. **Store Pavers:** Stack pavers in a designated area and allow them to continue curing if needed. Follow any additional manufacturer RECOMMENDATIONS.

❖ CURING FABRICATED CEMENT PAVERS

Curing fabricated cement pavers is a crucial process that ensures the pavers achieve their full strength and durability. Proper curing helps the cement hydrate fully, reducing the risk of cracks and improving the paver's overall performance.

Steps for Curing Fabricated Cement Pavers:

1. **Allow Initial Set:** After demolding, let the pavers remain in a shaded or covered area for the initial setting period, typically 24-48 hours.
2. **Maintain Moisture:**
 - **Covering:** Keep the pavers moist by covering them with wet burlap, damp cloths, or plastic sheeting. This helps retain moisture and prevent rapid drying.
 - **Watering:** Regularly mist the pavers with water if they are not covered, especially in hot or dry conditions. Ensure the surface remains damp but not soaked.
3. **Avoid Direct Sunlight:** Protect the pavers from direct sunlight and extreme temperatures, which can cause uneven drying and cracking. Use shading materials if necessary.
4. **Extended Curing Time:** Continue curing the pavers for at least 7-14 days. This duration may vary based on environmental conditions and specific mix requirements.
5. **Monitor Conditions:** Regularly check the pavers to ensure they are curing evenly. Adjust covering and watering as needed based on weather conditions.
6. **Prevent Damage:** Avoid heavy traffic or load-bearing activities on the pavers until they have fully cured. This helps prevent deformation and ensures proper strength development.
7. **Final Inspection:** After the curing period, inspect the pavers for any signs of damage or defects before installation.

❖ **Period of Curing:**

In Rwanda the minimum curing period 7days is a common minimum, extending the curing period to 14 days or even 28 days can contribute to further strength development and durability. Extended curing is especially beneficial in environments with challenging conditions, such as high temperatures or low humidity

IDENTIFICATION OF DRYING TYPES

❖ **Artificial drying**

Artificial drying of cement pavers is a controlled process used to accelerate the curing and drying of pavers, especially in situations where natural conditions are not ideal or when a quicker turnaround is needed.

Methods of artificial drying

- 1. Use of fans:** Circulating air around the pavers with fans can help accelerate the drying process. This method is relatively gentle and can be effective, especially when combined with careful monitoring.
- 2. Heating system:** Controlled heating systems can be used to increase the temperature in the drying environment. This method should be approached cautiously to prevent excessive drying and thermal cracking.
- 3. Dehumidifiers:** can be used to reduce the humidity in the drying environment. This method helps control the moisture content of the air and can contribute to the drying process.
- 4. Covered drying areas:** Construct covered areas with controlled environmental conditions for artificial drying. This approach provides protection from external factors and allows for better control over the drying process.
- 5. Ventilation:** Provide adequate ventilation to allow moisture to escape. Proper ventilation helps prevent the buildup of humidity, especially in enclosed drying spaces.

6. Monitoring and adjusting: Continuously monitor the moisture content, temperature, and humidity during artificial drying. Adjust the drying conditions as needed to maintain control and prevent issues.

Advantages of Artificial Drying:

- Accelerates the curing process for quicker production.
- Ensures consistent quality by managing temperature and humidity.
- Unaffected by external weather conditions, making it reliable in any climate.

❖ Natural drying

Natural drying in the context of cement pavers refers to the process of allowing the pavers to cure and dry naturally without the use of artificial methods like heat chambers, infrared heaters, or steam curing.

Steps for Natural Drying of Cement Pavers:

1. Initial Setting: After demolding, place the pavers in a shaded, well-ventilated area to begin the initial curing process, which typically lasts 24-48 hours.

2. Moisture Retention: To prevent rapid drying and surface cracking, cover the pavers with damp burlap, cloths, or plastic sheets. This helps retain moisture during the crucial early stages of curing.

3. Regular Watering: Periodically mist the pavers with water to maintain a consistent moisture level, especially in hot or dry conditions. This ensures the pavers don't dry out too quickly, which could lead to cracks.

4. Protection from Direct Sunlight: Shield the pavers from direct sunlight and extreme temperatures. Use shading materials or place the pavers in a naturally shaded area to avoid uneven drying.

5. Extended Curing Time: Allow the pavers to cure naturally over a period of 7-14 days or longer, depending on environmental conditions. During this time, the cement continues to hydrate, gaining strength.

6. Final Inspection: After the curing period, inspect the pavers for any signs of defects, such as cracks or surface imperfections.

7. Continued Curing (if needed): If required, continue to cure the pavers by keeping them moist for an extended period, ensuring maximum strength and durability.

Advantages of Natural Drying:

- Requires no additional equipment or energy, reducing costs.
- Curing often results in stronger, more durable pavers.
- Involves a straightforward process with minimal intervention.

Application of cement pavers curing process complying with Rwanda standards

In Rwanda, the application of the cement pavers curing process must comply with national standards to ensure the quality, strength, and durability of the pavers. The key aspects of the curing process align with good construction practices, which are essential for infrastructure projects like roads, walkways, and public spaces.

Application of Cement Pavers Curing Process Complying with Rwanda Standards:

1. **Compliance with Rwanda Standards:** Ensure that the curing process aligns with Rwandan building codes and standards, such as those outlined by the Rwanda Bureau of Standards (RBS) or other relevant authorities. This includes adherence to specified curing times, methods, and quality control measures.
2. **Initial Curing Period:** After Demolding, the pavers should be placed in a controlled environment where they are protected from direct sunlight, wind, and rain. According to Rwandan standards, the initial curing period typically lasts 24-48 hours.
3. **Moisture Retention:** Cover the pavers with damp materials like burlap or plastic sheets during the curing period to retain moisture. This step is crucial to prevent premature drying, which can lead to cracking and reduced strength, ensuring the pavers meet the durability requirements set by local standards.
4. **Extended Curing Duration:** The pavers should be cured for at least 7-14 days, with regular watering or misting to maintain moisture. Rwanda's standards emphasize the importance of adequate curing time to achieve the required compressive strength.
5. **Environmental Considerations:** The curing environment should comply with Rwanda's climate conditions, taking into account temperature and humidity. For instance, in Rwanda's warm climate, preventing rapid drying is critical, and thus proper shading and moisture retention are essential.
6. **Quality Control and Testing:** Regular testing of the pavers during and after curing is necessary to ensure they meet the required strength and durability as per Rwanda standards. This includes compressive strength tests to verify that the pavers can withstand local usage conditions.

7. **Documentation and Compliance:** Maintain records of the curing process, including methods used duration, and environmental conditions. This documentation may be required for compliance checks by Rwandan regulatory bodies.
8. **Post-Curing Handling:** After the curing process, handle the pavers carefully to avoid damage. Store them properly, ensuring they are protected until installation.

LEARNING OUTCOME 4: PREPARE GROUND BASE

❖ Preparation of working area

Methods of assessing the paving area

1. Site Surveying:

- **Purpose:** To obtain precise measurements and understand the layout of the area.
- **Tools Used:** Total stations, theodolites, GPS devices, or laser levels.
- **Outcome:** Accurate dimensions, elevations, and boundaries of the paving area.

2. Manual Measurement:

- **Purpose:** To measure the size and shape of the area manually.
- **Tools Used:** Measuring tapes, measuring wheels, and stakes.
- **Outcome:** Basic measurements for straightforward or small-scale projects.

3. Topographical Survey:

- **Purpose:** To assess the terrain and elevation variations.
- **Tools Used:** GPS, total stations, or drones.
- **Outcome:** Detailed mapping of slopes, contours, and drainage patterns.

4. Soil Testing:

- **Purpose:** To evaluate the soil's suitability for paving, including its strength and drainage capacity.
- **Tools Used:** Soil augers, penetrometers, and laboratory analysis.
- **Outcome:** Data on soil stability and recommendations for base preparation.

❖ Clearing and cleaning methods of the paving area

Clearing and cleaning the paving area involves preparing the site by removing obstacles, debris, and any other materials that could interfere with the paving process. This preparation is crucial for ensuring a stable base and a smooth installation of the paving materials.

❖ **Clearing Methods:**

1. Vegetation Removal: Cut down and remove all vegetation, including the roots, to prevent future growth that could damage the paving.

- **Purpose:** To eliminate grass, shrubs, trees, roots, and other plant life from the paving area.
- **Tools Used:** Shovels, spades, axes, brush cutters, or heavy machinery like bulldozers for larger vegetation.

2. Debris and Obstacle Removal: Collect and remove all debris and obstacles to create a clean, level surface.

- **Purpose:** To clear the site of any debris, rocks, old pavement, or other obstacles.
- **Tools Used:** Shovels, rakes, wheelbarrows, and in some cases, excavators or bulldozers.

3. Excavation: Excavate to the depth needed for the base and sub-base layers, ensuring the area is ready for leveling and compaction.

- **Purpose:** To remove excess soil or to dig out the area to the required depth for the paving installation.
- **Tools Used:** Excavators, backhoes, or manual digging tools.

❖ **Cleaning Methods:**

1. Surface Cleaning: Sweep or blow away loose materials to ensure a clean surface for base preparation.

- **Purpose:** To remove loose dirt, dust, and small debris from the cleared area.
- **Tools Used:** Brooms, blowers, pressure washers, or vacuum sweepers.

2. Soil Compaction: Compact the soil to reduce voids and increase its load-bearing capacity, providing a firm foundation for the paving materials

- **Purpose:** To compact the soil to create a solid and stable base for paving.
- **Tools Used:** Plate compactors, rollers, or rammers.

3. Edge Trimming: Trim the edges of the paving area to create a clean, well-defined boundary.

- **Purpose:** To define the boundaries of the paving area and remove any excess soil or vegetation along the edges.
- **Tools Used:** Edging tools, spades, or trimmers.

❖ **Types of waste on site**

Here are the types of waste :

1. Biodegradable Waste: Waste that can be broken down naturally by microorganisms into harmless substances.

- **Examples:** Food scraps, yard waste (leaves, grass clippings), paper, and organic materials.
- **Disposal/Management:** Typically composted or processed through anaerobic digestion to create compost, biogas, or fertilizers.

2. Chemical Waste: Waste that contains harmful chemicals, which may be hazardous to human health and the environment.

- **Examples:** Pesticides, solvents, industrial chemicals, and laboratory waste.
- **Disposal/Management:** Requires special handling and disposal methods, such as incineration, chemical treatment, or containment in hazardous waste facilities.

3. Demolition Waste: Waste generated from the demolition of buildings and other structures.

- **Examples:** Concrete, bricks, wood, metal, and glass.
- **Disposal/Management:** Often recycled or reused, with materials like concrete and metal being processed for new construction projects, while non-recyclable materials are sent to landfills.

4. E-Waste (Electronic Waste): Waste consisting of discarded electronic devices and components.

- **Examples:** Old computers, smartphones, televisions, and batteries.
- **Disposal/Management:** Requires specialized recycling processes to recover valuable materials like metals and to safely dispose of toxic substances like lead and mercury.

5. Industrial Waste: Waste generated from industrial activities, including manufacturing, mining, and production processes.

- **Examples:** Slag, ash, chemical by-products, and scrap materials.
- **Disposal/Management:** Depending on its composition, industrial waste may be treated, recycled, or disposed of in landfills designed to handle industrial materials.

6. Recyclable Waste: Waste materials that can be processed and used again to create new products, reducing the need for raw materials.

- **Examples:** Paper, cardboard, glass, metals, and certain plastics.
- **Disposal/Management:** Collection, sorting, and processing at recycling facilities, where the materials are transformed into new products or raw materials for manufacturing.

❖ Investigation of ground base moisture.

➤ Conduct soil investigation

Conducting a soil investigation refers to the process of examining the physical and chemical properties of soil at a specific site to determine its suitability for construction or other purposes.

Function of conducting a soil investigation

- Determine if the soil can support the planned structure.
- Provide data for designing appropriate foundations.
- Detect problems like erosion, liquefaction, or instability.
- Assess how much load the soil can safely support.
- Understand how construction may affect soil and groundwater.
- Help estimate construction costs and plan for potential soil improvements.

Information to be gathered during soil investigation

During a soil investigation before paving an area, the following information should be gathered:

- **Soil Composition:** Identify the types of soil present (e.g., clay, sand, silt) and their distribution.
- **Soil Strength:** Measure the shear strength and load-bearing capacity to ensure the soil can support the pavement.

- **Moisture Content:** Determine the soil's moisture levels, as they affect compaction and stability.
- **Soil Density:** Assess the compaction level to ensure the soil is properly compacted to support the pavement.
- **Drainage Characteristics:** Evaluate how well the soil drains water to prevent pooling or erosion under the pavement.
- **Soil pH and Chemical Properties:** Test for acidity, alkalinity, or contaminants that could affect the pavement materials.
- **Compaction Characteristics:** Understand how the soil will compact under load, which affects the pavement's durability.
- **Subsurface Conditions:** Identify any underground structures, utilities, or water tables that could impact paving.

❖ **Drainage system application during Investigation of ground base moisture**

During the investigation of ground base moisture, a drainage system is applied to manage and mitigate excess moisture in the soil. This application involves:

1. **Identifying Moisture Sources:** Determine the sources of moisture, such as groundwater, surface water, or infiltration, that may affect the soil beneath the planned pavement area.
2. **Assessing Soil Permeability:** Evaluate how well the soil drains water naturally. Soils with low permeability may require additional drainage solutions.
3. **Designing Drainage Solutions:** Based on the moisture levels and soil permeability, design a drainage system to direct excess water away from the pavement area.
4. **Preventing Water Accumulation:** Ensure that the drainage system effectively prevents water from pooling under the pavement, which could weaken the soil and lead to pavement failure.

❖ **Watering of ground base during Investigation of ground base moisture**

Watering the ground base moisture is a controlled procedure used to assess how the soil reacts to added moisture.

Here's how it's typically applied:

1. **Simulating Rainfall:** Water is applied to the ground base to simulate rainfall or other moisture conditions. This helps to observe how the soil absorbs, retains, and drains water.
2. **Measuring Infiltration Rate:** The rate at which water infiltrates into the soil is measured. This information is critical to understanding the soil's permeability and drainage characteristics.

3. **Evaluating Soil Swelling or Shrinkage:** Some soils, especially those with high clay content, may swell when wet and shrink when dry. Watering helps assess these potential changes in volume.
4. **Observing Surface Water Behavior:** The process allows for observation of how water moves across the surface, identifying any areas where water might pool or run off, which can impact the pavement.
5. **Testing Compaction Response:** Watering can also be used to evaluate how well the soil compacts when moisture is present. This helps in planning the compaction process during construction.
6. **Determining Optimal Moisture Content:** The procedure helps to find the optimal moisture content for soil compaction, which is crucial for achieving a stable base for paving.
7. **Assessing Drainage Needs:** The results of watering help determine if additional drainage systems are necessary to prevent water accumulation under the pavement.

❖ **Excavation and levelling of the area to be paved**

✓ **Excavation**

Excavation refers to the process of removing earth, soil, rock, or other materials from a site to create a cavity or hole in the ground.

Methods of excavation

- **Manual Excavation:** Manual excavation involves the removal of soil, rock, or other materials by hand using basic tools or hand tools such as hoe, spade, shovel etc.
- **Mechanical Excavation:** Mechanical excavation involves the use of machinery to remove large volumes of earth, rock, or other materials backhoe loader, excavator, and bulldozer.

Function of excavation before paving area

1. Ensures a solid foundation by removing unsuitable material.
2. Creates a flat surface for even pavement.
3. Allows space for gravel or other supportive layers.
4. Facilitates proper water drainage to prevent pooling.
5. Removes rocks, roots, and debris that could affect the pavement.
6. Sets the right slope for water runoff.
7. Enables thorough compaction to prevent settling.

✓ **Techniques of levelling working area**

Excavating and leveling an area before paving is a critical step in ensuring a stable and even surface for your construction project. Here are the steps and techniques for excavation and leveling:

1. Excavation:

- **Marking and Layout:** Begin by marking the boundaries of the area to be excavated using stakes and strings or marking paint. Ensure the layout is in accordance with the design and dimensions of the paved area.
- **Clearing and Vegetation Removal:** Remove any vegetation, trees, or debris from the area.
- **Equipment Selection:** Choose the appropriate excavation equipment based on the size and depth of the excavation. Common equipment includes **excavators**, backhoes, and bulldozers for large projects.
- **Depth Control:** Determine the desired depth of excavation and set the equipment accordingly. Use a laser level or GPS equipment for precise depth control.
- **Gradual Excavation:** Excavate the area gradually, starting from the perimeter and working toward the center. Remove soil in layers, checking for any unexpected obstacles or utilities.
- **Soil Disposal:** Dispose of excavated soil properly, adhering to local regulations.

2. Leveling the Working Area:

- a. **Rough Grading:** After excavation, perform rough grading to achieve the initial levelness of the area.
 - b. **Compaction:** Compact the soil thoroughly to ensure stability and prevent settling. Use compaction equipment such as rollers or plate compactors.
 - c. **Fine Grading:** After rough grading and compaction, perform fine grading using a grader or a similar piece of equipment to ensure the surface is even and smooth, ready for paving.
 - d. **Laser Level or String Lines:** Use laser levels or string lines to achieve precise grade control during fine grading. Measure and adjust the elevation to meet the required specifications.
 - e. **Quality Control:** Continuously monitor and measure the surface elevation during the leveling process to ensure it meets the design requirements.
- Surface Inspection:
- f. **Verification and Final Inspection:** Before paving, conduct a final inspection and verification of the area to ensure it meets the specified grade and slope requirements.

3. Creating your reference

Put the edges at the area, wind strings around them at ground level and tie them tightly and use those strings to create your leveling reference.



4. Leveling the strings and adding the slope

- The strings will mark the height at which the surface of your pavers will be in the future, they will be our reference for digging and everything else from now on. That said, it's in your best interest to level them properly.
- Using the carpenter's level ensure the strings are leveled all the way. Then, in order to allow for drainage, create a slope of 1 inch to every 4 feet of length. This slope should be added only in the direction you want the water to flow to.



5. Digging

- First, calculate what's the total depth of your paving project after deciding on how much gravel and sand to add. In order to find out how much you will need to dig, you need to add that up to the depth of the pavers.
- Now, before you start digging, you should also consider that some edging will be necessary to hold everything in place using the strings as reference, dig the soil to the necessary depth. Don't forget to pack down the soil with the plate compactor.



6. Add crushed gravel

Drive stakes whose height will represent the limit for the gravel through the area. Remember all height references should be measured from the string and then down spread the crushed gravel. Ensure the gravel is properly leveled by running a straight lumber through the whole area, always making sure the gravel is at the same height of the stakes.

7. Add sand

In a procedure that's similar to that of the last step, drive stakes through the area whose height will represent the total height of sand you will be adding. Again, level those stakes according to the reference strings and considering the fact that you want to fill the area with sand in a way that the pavers' height will be the only thing left to add. spread the sand through the whole area and level it with the lumber making sure it is at the same height as the stakes. **Next, you need to install paver restrains and the pavers themselves.**

❖ Compaction of working area

Compaction of a working area refers to the process of compressing the soil or base material to increase its density and stability.

✓ Advantages of compacting ground base

- Increased Load-Bearing Capacity
- Prevents Settlement
- Improved Stability
- Enhanced Durability
- Better Water Drainage

✓ Methods of compacting ground base

1. Manual method

A compaction technique where human labor is used to compress the soil or base material without the aid of motorized equipment.

- **Hand Tamper:** A simple tool with a heavy, flat base and a handle. The user manually lifts and drops the tamper to compact small areas, typically used for minor projects or hard-to-reach places.
- **Ramming:** Involves using a rammer or a rod to manually compress the soil. Suitable for compacting soil in tight spaces or around edges where machinery can't reach.

2. Mechanical method

A compaction technique that utilizes motorized equipment to compress the soil or base material. Mechanical methods include the use of machines such as plate compactors, roller compactors, vibratory compactors, and other specialized equipment.

- **Plate Compactor:** A motorized device with a heavy, flat metal plate that vibrates to compact the soil. Ideal for compacting granular soils (like sand or gravel) and small to medium-sized areas.
- **Roller Compactor:** Large machines with heavy rollers (smooth or pad-footed) used for compacting large areas. Commonly used in road construction and large paving projects.
- **Vibratory Compactor:** Uses vibrations to compact the soil, often attached to excavators or as standalone machines. Effective for both granular and cohesive soils.
- **Pneumatic Tyred Roller:** A machine with rubber tires that apply pressure to the soil. The tires can be adjusted for different pressures, making it versatile for various soil types.

LEARNING OUTCOME 5: LAY PAVERS

Construction pavers are flat, durable blocks made from materials like concrete, stone, or brick, used to create hard surfaces such as walkways, driveways, patios, and roads.

❖ Types of pavers

- **Cement Pavers:** These are made from a mixture of cement and aggregates. Also used for driveways, patios, and walkways.
- **Concrete Pavers:** Similar to cement pavers, concrete pavers are made from a mixture of cement, aggregates, and coloring agents. They come in various

shapes, sizes, and colors, making them versatile for various landscaping projects.

- **Brick Pavers:** Made from fired clay, brick pavers have a timeless and classic appearance. They are commonly used for pathways, driveways, and outdoor patios.
- **Cobblestone Pavers:** Cobblestones are natural stones, typically granite or basalt, which are cut and shaped for use as pavers. They give a rustic and charming look to driveways and streets.
- **Rubber Pavers:** These pavers are made from recycled rubber tires and are known for their shock-absorbing qualities. They are often used in playgrounds, sports courts, and areas where impact resistance is important.
- **Bluestone Pavers:** Bluestone is a type of sandstone that is known for its blue-gray color. It's a popular choice for creating elegant and natural-looking patios and walkways.
- **Flagstone Pavers:** Flagstone is a flat and thin sedimentary rock often used for paving.
- **Travertine Pavers:** Travertine is a type of limestone formed by mineral deposits from natural springs. It has a unique texture and is often used for pool decks, patios, and outdoor entertainment areas.

❖ **Methods/processes of Setting out the area to be paved**

"**Setting out the area to be paved**" refers to the process of preparing and marking the ground where paving will take place.

It involves:

- Defining Boundaries
- Leveling
- Marking Layout

1. Use of Calibrated Measuring Tools and Calibrated Levelling Tools

✓ **Measure and Mark Boundaries:**

- Use calibrated measuring tapes or laser distance meters to accurately measure the dimensions of the area to be paved.
- Mark the boundaries with stakes, string lines, or chalk to define the perimeter of the paved area.

✓ **Establish Reference Points:**

- Use a calibrated level (e.g., spirit level, laser level) to set up reference points at consistent heights across the area. These reference points help ensure

that the surface remains even and slopes as required for drainage.

2. Mark Out the Level of Pavement

- ✓ **Set a Benchmark:** Establish a fixed point (benchmark) at the desired pavement level. This will serve as a reference for all other points.
- ✓ **Create Guide Lines:**
 - Stretch string lines across the area at the height of the finished pavement level, using the benchmark as a reference.
 - Use a spirit level or laser level to ensure the string lines are perfectly level or follow the necessary slope for drainage.
- ✓ **Check and Adjust:**
 - Regularly check the level of the ground relative to the string lines. Adjust the height of the base material as needed to match the desired level, ensuring consistency across the entire area.

Methods and processes involved in setting out the area to be paved, including the use of calibrated measuring and leveling tools:

- 1. Site Inspection and Preparation:** Begin by conducting a thorough site inspection to assess the condition of the ground and existing features. Remove any obstacles, debris, or vegetation from the area to be paved.
- 2. Define the Layout:** Determine the desired layout and dimensions of the paved area. This can be based on architectural plans, design drawings, or project specifications.
- 3. Measurements and Calibration:** Accurate measurements are crucial. Use calibrated measuring tools such as measuring tapes, laser distance meters, or measuring wheels.
- 4. Leveling Tools:** Use calibrated leveling tools such as a builder's level, laser level, or a theodolite to establish the desired slope, if applicable (e.g., for drainage purposes).
- 5. Establish Reference Points:** Set up benchmarks or reference points around the perimeter of the project area. These reference points will help maintain consistency during the paving process.
- 6. String Lines and Profiles:** String lines or profiles are used to maintain the correct height and alignment during paving. These should be set at the desired

elevation and slope.

7. Marking the Level of Pavement: Using the calibrated leveling tools, mark the desired level or slope of the pavement at regular intervals along the string lines or profiles.

8. Final Checks: Double-check all measurements, elevations, and reference points before proceeding with paving.

By following these steps and using calibrated measuring and leveling tools, you can ensure that the area to be paved is accurately set out, resulting in a high-quality finished pavement.

❖ **Supporting the edge of area to be paved**

"**Supporting the edge of the area to be paved**" refers to stabilizing and reinforcing the borders or perimeter of the paved area to ensure it remains intact and does not shift or degrade over time.

✓ **Functions of supporting the edge of the area to be paved**

1. Keeps the paving materials from moving or slipping out of place.
2. Maintains Shape
3. Enhances Stability
4. Reduces Wear
5. Improves Drainage

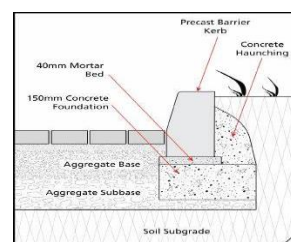
✓ **Methods of Supporting the edge**

Various methods can be used to provide this support while complying with related standards in Rwanda. Three common methods are:

- **Precast Concrete Kerbs:**

Precast concrete kerbs are a popular choice for supporting the edge of paved areas. They are durable, long-lasting, and readily available in various sizes and shapes.

Description: Preformed concrete units designed to form a continuous edge along the paving area.



- **Cut Stone:**

Natural or cut stone can be used as an edge restraint for paved areas, particularly in landscaping and architectural applications.

Description: Natural stone blocks cut to size and used as edge restraints.



- **Plastic Edge Restraints:**

Plastic edge restraints are lightweight and easy to handle. They are suitable for a range of applications and are often used in residential.

Description: Edge restraint systems made from flexible or rigid plastic materials.

- ❖ **Spreading the sand bedding along the ground base.**

"**Spreading the sand bedding along the ground base**" refers to the process of laying a layer of sand on the prepared base before installing paving materials.

Importances of spreading the sand bedding along the ground base

- Provides a smooth, even surface for paving materials.
- Acts as a buffer to distribute loads and reduce stress on the base.
- Allows for minor adjustments to ensure proper alignment of paving materials.
- Helps to stabilize the paving materials, reducing the risk of shifting or settling.

- ✓ **Identification of the Type of Sand to Be Used**

- **Coarse Sand Specification Complying with Rwanda Standards:**

- **Particle Size:** Coarse sand typically has particles ranging from 0.5 mm to 25 mm. This specification ensures that the sand provides adequate drainage and support.

- **Rwanda Standards:** Compliance with Rwanda Standards ensures that the sand is suitable for use in construction and paving, meeting criteria for particle size, cleanliness, and absence of harmful impurities.
- **Advantages:** Coarse sand allows for good water drainage, which helps prevent water logging and erosion. It also compacts well, providing a stable base for paving materials.

✓ **Determination of Thickness of Sand Layer Applied**

Guideline for Thickness:

The thickness of the sand layer typically ranges from 1 to 2 inches (25 to 50 mm). This provides a sufficient cushion and leveling surface for the paving materials.

➤ **Factors Affecting Thickness:**

- Base Condition
- Paving Material
- Load Requirements

➤ **Application Process:**

- **Spreading Sand:** Spread the coarse sand evenly over the prepared base. Use a rake or similar tool to distribute the sand uniformly.
- **Leveling:** Use a screed or leveling tool to ensure the sand layer is consistent in thickness and perfectly flat.
- **Compacting:** Lightly compact the sand to remove any air pockets and create a stable, firm surface for the paving materials.

❖ **Laying of Pavers on Sand Bedding Surface**

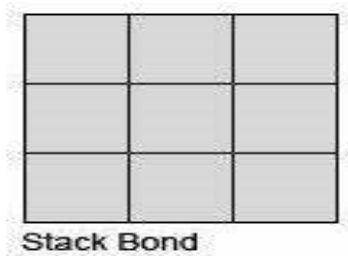
➤ **Identification of Pattern**

Pattern refers to the specific arrangement or design of the pavers as they are laid out on the surface.

1. Stack Bond: Pavers are laid in straight rows, with each row aligned directly above the one below it, creating a grid-like pattern.

- **Advantages:** Simple and easy to install; creates a clean, uniform appearance. Suitable for modern or minimalist designs.
- **Disadvantages:** Limited in aesthetic variety and may be less stable under heavy load conditions compared to other patterns.

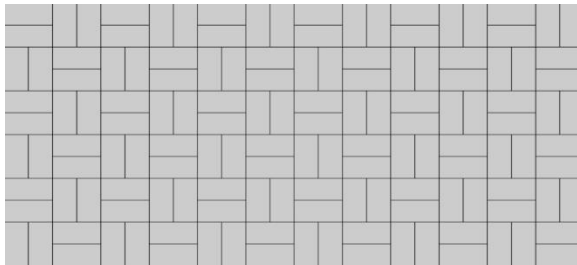
Stack Bond: Simple, uniform; suitable for modern designs.



2. Basket Weave Bond: Pavers are arranged in a pattern where pairs of pavers are laid in alternating vertical and horizontal directions, creating a woven effect.

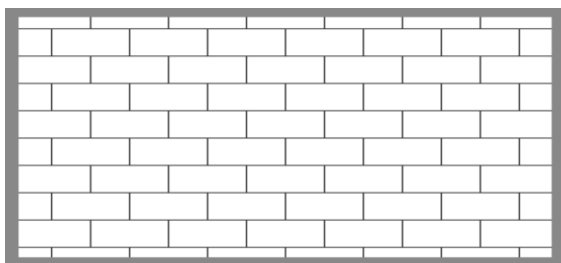
- **Advantages:** Provides a textured, interlocking appearance that can enhance the aesthetic appeal. Effective at distributing load and providing stability.
- **Disadvantages:** More complex to install; requires careful alignment and planning to ensure uniformity.

Basket Weave Bond: Decorative, interlocking; enhances stability and aesthetic appeal.



3. Running Bond Pavers are laid in staggered rows, where each row is offset from the one below it, creating a brick-like appearance.

- **Advantages:** Enhances strength and stability by overlapping joints; creates a visually appealing, classic pattern.
- **Disadvantages:** Requires precise measurement and cutting, especially on edges.
- **Running Bond:** Staggered, classic; offers improved strength and visual interest.



❖ Pavers Laying Method

1. Traditional Pavers Laying Method:

This is the most common method for installing pavers. It involves preparing a base of compacted gravel or sand, which provides a stable foundation. The pavers are then placed on top of this base, and typically, they are laid in a specific pattern or design. After the pavers are in place, sand is spread over them and swept into the joints to lock them in position. This method creates a durable and stable surface suitable for driveways, walkways, and patios.

2. Ridged Laying Method

Ridged Laying Method: The ridged laying method is a variation of the traditional paver laying method. Instead of using a smooth layer of sand to bed the pavers, this method involves using ridged or grooved spacers to create small gaps between the pavers as they are installed. These gaps can be filled with sand or a similar material to provide a more stable surface with improved interlock between the pavers. The ridged method can enhance the strength and stability of the paver installation.

3. Flexible Laying Method

Flexible Laying Method: The flexible laying method, often referred to as "flexible paving," is a technique used for pavers on a flexible base, such as grass, soil, or gravel. It's commonly used in applications where a rigid base like concrete isn't desired. In this method, pavers are placed directly on the flexible surface without the need for a concrete or gravel base. The gaps between the pavers can be filled with soil or left open to allow for grass to grow, creating a permeable surface. This approach is often used in areas like grass driveways or pathways where a more natural look is desired.

Construct pavers complying with Rwandan Standard: precast concrete paving blocks specification.

10-Step Guide to Installing Pavers

Whether you plan to install brick, concrete or stone pavers, the following paver installation steps should help. Even if you plan to have your pavers professionally installed it is a good idea to be familiar with the process.

1. Planning and Layout

2 Calculating Amount of Pavers Needed

3. Excavation
4. Base Material
5. Edge Restraints
6. Sand Bedding
7. Laying Pavers
8. Sand Joints
9. Sealing
10. Maintenance

1. Planning and Layout

- Planning the area to be paved is your first step. Decide where your pavers will go and determine if there is enough space for them. Your plan can be drawn out and measured using graph paper.
- Lay out your planned area using a hose or some wood to form the outline of your paved area. Next, mark or spray paint the outline, approximately eight inches outside that outline.
- Remember that the area needs to have gradient or slope, so water can run off the pavers to prevent pooling or ponding that can damage the pavement over the long term.

2. Calculating Amount of Pavers Needed

- First measure the width and the length of the area to be paved. Then multiply the width by the length. The result gives you the square footage of the area to be paved.
- You should always add 5% to this figure, to allow for the pavers that will be cut to fit curves and odd spaces. If you know that you will be cutting a lot of pavers, you should add 15%.
- The square footage of the area to be paved should then be multiplied by the paver coverage rate which can be obtained from your supplier. This rate varies with the size and shape of the paver to be used.

Quantities of base course material and bedding sand are calculated in volume of cubic metres (m^3 – i.e. length x width x depth). You will need to allow a bulking factor for bedding sand and base course material so that you achieve correct depth once they are compacted. The Supplier can tell you what percentage to allow.

Paver quantities are calculated in area of square metres (m^2 – i.e. length x width). Divide the total area by the area of a single paver to work out the number of pavers you need. If any pavers need to be cut because of the laying pattern, allow for some wastage and order a few extra pavers.

Example:

Calculate the volume of materials required for the following:

Area to be paved measures 5.4 m x 2.1 m

Base course laid to depth of 150mm

Bedding sand lay to depth of 30mm

Pavers measure 150 x 150 x 40mm

Total depth of excavation = combined height of bedding sand, base course and paver (150 + 30 + 40mm) = 220mm.

Calculations

Base course (m³) = (l x w x d) + bulking factor

= (5.400 x 2.100 x 0.150) + 15%

= 1.701 + 0.255

= **1.956m³**

Bedding sand (m³) = (l x w x d) + bulking factor

= (5.400 x 2.100 x 0.030) + 5%

= 0.340 + 0.017

= **0.357m³**

Pavers (total needed) = total area ÷ paver area

= (5.400 x 2.100) ÷ 0.150 x 0.150

= 11.340 ÷ 0.0225

= **504**

3. Excavation

Establish the height that the pavers should reach and not exceed.

Calculating the amount of excavated material

- Depending on the area being paved, the amount of material excavated can be significant. It then needs to be removed from the site either by wheelbarrow to another area close by or by filling a skip or bin which is then taken away by truck.
- As with all compact material that is loosened, the actual volume can increase by varying amounts, depending on: the method used to excavate the type of material being excavated the moisture content of the material.
- This is called the bulking factor. Soil has a bulking factor of between 10 to 30%.

Calculate the volume of excavated material, plus a bulking factor of 20%, for an area measuring 5.4 m x 2.1 m with a total depth of excavation of 220mm.

Calculation

Volume = length x width x height

= 5.400 x 2.100 x 0.220

= 2.494m³

Bulking factor = 2.494 x 20%

= 0.498

Excavated material (volume) = 2.494 + 0.498

= 2.992m³

Approximately 3m³ to be excavated.

4. Base Material

Once the excavation is complete, the next step is the laying of the base material. The base material usually consists of crushed rock, such as 3/4 minus gravel.

5. Edge Restraints

Edging is essential in the maintenance of your paved area. If edge restraints are not installed, your pavers will move and separate. Harsh weather and high traffic usage, along with no edge restraints, will ruin your paved surface. Edge restraints can be made of plastic, pre-cast concrete, metal or w Install secure edge restraints before laying and screeding the bedding course and paving units.



6. Sand bedding

Before laying the pavers, a layer of bedding sand is placed over the compacted base material. This layer provides a bed into which the pavers are set. The sand bedding also helps to protect the sand joints from being eroded away. Lay down one inch diameter PVC pipe across the base material. Next spread a layer of sand between one and one and a half inches thick. Now, use a 2 x 4 to screed across the sand so the PVC piping is visible. Carefully remove the PVC pipe.



7. Laying Pavers

Lay the pavers in the pattern and design of that you have planned for. Lay

them close together. Some pavers have space bumps incorporated into the paver design. If pavers need to be cut to size, use a diamond saw, masonry chisel or mechanical splitter and safety glasses.



After all the pavers are laid, use a mechanical plate compactor to help set the pavers into the sand bedding.

8. Sand Joints

This step involves sweeping sand over the surface which fills up the joints locks the pavers into place.

It is important to use finegrained sand rather than a coarse one. Sifted and bagged play sand works well as does that labeled as “paver sand”. The reason for this is that fine sand will compact much better and fill up joints quicker. The sand and paver area must be dry to complete this step.



There are two main types of joints used for this purpose:

- **Mortar Joint:** A mortar joint is created by filling the gaps between pavers with a mixture of cement, sand, and water. This mixture forms a strong and durable bond between the pavers, effectively locking them in place.
- **Sand Joint:** Sand joints, as the name suggests, involve filling the gaps between pavers with fine-grained sand. Polymeric sand, which contains additives that harden when exposed to water, is commonly used for this purpose.

9. Sealing

First, clean and sweep the newly paved area, in order to remove all materials and debris.

Seal the area with a sand binding sealant, this will ensure that the joint sand

will not disappear. This will also prevent the growth of vegetation between the joints. Sealing your pavers helps prevent stains from oil, grease, grime, rust, moss, algae, tire markings and day-to-day spills. The sealant allows you to clean off such stains with greater ease.

❖ **Types of paver sealers:**

1. Solvent-Based Paver Sealer:

Solvent-based sealers are made with petroleum-based solvents. They provide a durable and long-lasting seal that enhances the color of the pavers.

2. Water-Based Paver Sealer:

Water-based sealers are environmentally friendly and have lower VOC (volatile organic compound) levels compared to solvent-based sealers.

3. Wet Look Paver Sealer:

Wet look sealers are designed to enhance the appearance of your pavers by giving them a wet or glossy look.

4. Gloss Paver Sealer:

Gloss sealers provide a high-shine finish to your pavers, giving them a polished and reflective surface.

❖ **Properties of sealer materials**

- **Elasticity:** Sealants need to be elastic to accommodate movement in the materials they are sealing. They should be able to stretch and compress without losing their integrity.
- **Adhesion:** Sealants should have good adhesion properties to bond effectively to the surfaces they are sealing. Strong adhesion helps prevent leaks and maintains the integrity of the seal.
- **Modulus:** Modulus refers to the stiffness or rigidity of a sealant material. The modulus should be selected based on the specific application requirements.
- **Compatibility:** Sealants must be compatible with the materials they are in contact with, such as substrates, paints, and coatings.
- **Weatherability:** Sealants are often exposed to various environmental conditions, including UV radiation, moisture, temperature fluctuations, and chemical exposure.
- **Cohesion:** Cohesion is the internal strength of the sealant material. It measures the ability of the sealant to hold together as a cohesive mass, maintaining its structural integrity.

- **Chemical Resistance:** Depending on the application, sealants may need to resist exposure to chemicals, oils, solvents, and other substances.
- **Cure Time:** Sealants typically have a specific cure or drying time.
- **Durability:** The overall durability of sealant materials is crucial, as they are expected to provide long-term protection and sealing.

10. Maintenance

Maintenance is usually not an issue when it comes to pavers, if they have been laid properly. The overall function and use of your paved surfaces should be stable.

Re-sealing your pavers may be necessary on a regular basis to maintain color and appearance. It also helps protect the pavers against stains, algae and loss of joint sand.



❖ Cleaning work area, tools and equipment

Cleaning work area, tools and equipment refers to the process of removing all of the unwanted things to the work area, tools and equipments.

Methods of cleaning tools and equipments

1. Manual Clearing Method:

Manual clearing involves using hand tools and physical effort to clean and organize the work area, tools, and equipment. **Here are some common steps and techniques for manual clearing:**

- Remove Debris
- Organize Tools
- Wipe Down Surfaces
- Inspect Equipment
- Dispose of Waste
- Safely Store Materials

2. Mechanized Clearing Method:

Mechanized clearing involves the use of machines and specialized equipment to clean and organize the work area, tools, and equipment. **Here are some common steps and techniques for mechanized clearing:**

- Use Cleaning Machines: Employ industrial cleaning machines such as floor scrubbers, sweepers, and pressure washers to clean large work areas efficiently.
- Vacuum Systems.
- Automated Tool Storage.

❖ **Storage of tools and equipment**

➤ **Methods of storing tools and Equipment**

When storing tools and equipment used in a paving area, it's crucial to ensure that they are organized, protected, and easily accessible. Here are some specific methods tailored to the needs of a paving project:

1. Dedicated Storage Sheds or Containers

- **On-Site Sheds:** Set up a weatherproof shed or container close to the paving site. This provides a secure, organized space for storing tools and equipment during and after the project.

2. Tool Racks and Hangers

- **Wall-Mounted Racks:** Install heavy-duty racks or hangers on the walls of storage sheds for hanging tools like shovels, rakes, and brooms. This keeps them off the ground, organized, and easily accessible.

3. Heavy Equipment Storage

- **Designated Parking Areas:** Create a designated area for parking heavy machinery like pavers, rollers, and compactors. Use durable tarps or covers to protect equipment from the elements when not in use.

4. Tool Organization Systems

- **Tool Belts and Bags:** Keep frequently used smaller tools like trowels, levels, and measuring tapes in tool belts or bags. These can be hung on hooks or stored on shelves when not in use.